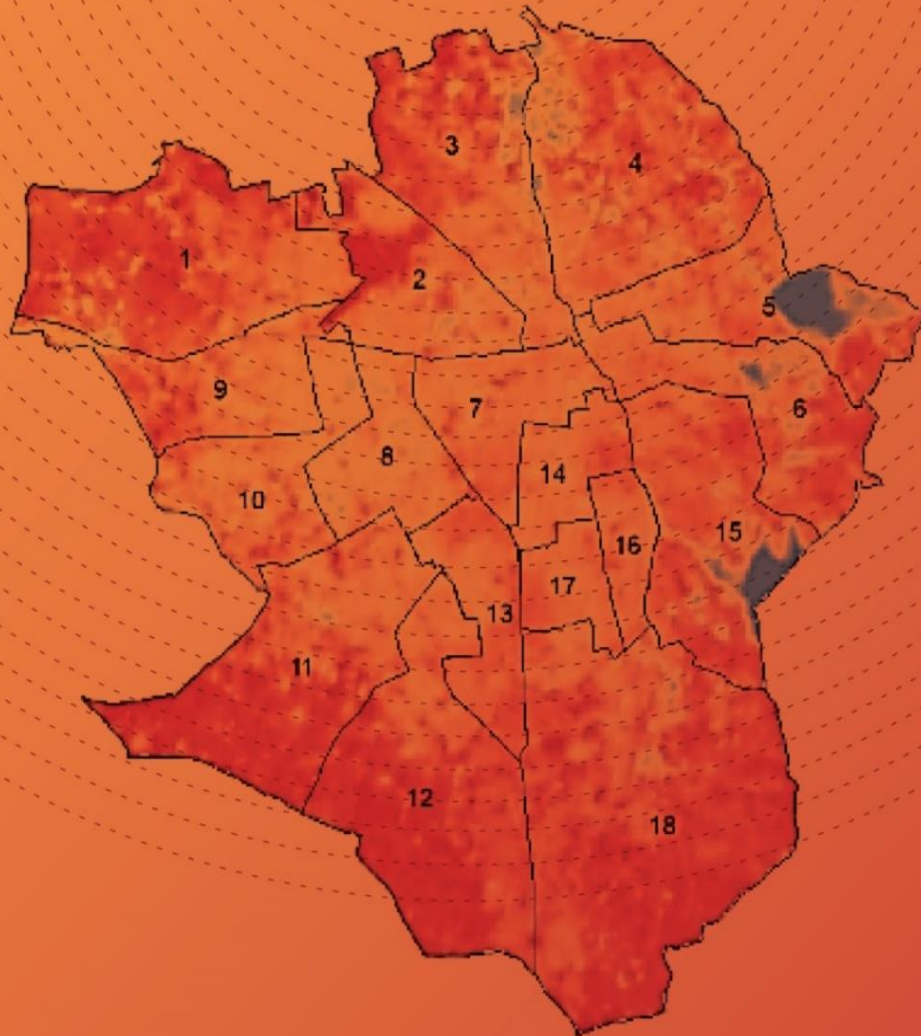




HEAT WAVE ACTION PLAN RAJKOT CITY 2024



Prepared by:

Integrated Research and Action for Development



Indian Institute of Public Health, Gandhinagar



Supported by:

Rajkot Municipal Corporation



International Development Research Centre, Government of Canada (IDRC)



Canada



India Meteorology Department

‘Heat Wave Action Plan- Rajkot City’

Supported by: International Development Research Centre, Government of Canada (IDRC)



Canada

Heat Wave Action Plan - Rajkot city, Gujarat

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1. Background and context

As the globe continues to warm up, it sets a new heat record every year across the world. As per the Intergovernmental Panel on Climate Change (IPCC)¹, hot days, hot nights and heat waves have become more frequent over the last 50 years. IPCC is a UN body that regularly releases scientific assessments on human-induced climate change.

In its sixth assessment report the UN intergovernmental panel stated that it is likely that surface temperatures over South Asia will increase more than the global average, with a projected rise of 4.6 °C (3.4 °C-6.0 °C) during 2081-2100 (IPCC AR6, 2020). This will have a devastating impact on water supply, electricity, sanitation, and food. Studies show that low and middle-income countries are most likely to bear a very high burden of deaths (World Health Organization, 2014)². The combination of rising temperatures and a large vulnerable population will worsen the impact of heat waves on developing countries.

In India, there is a well-recognized association between heat waves and morbidity and mortality. There is evidence of a threshold at 40 °C above which mortality increases³. According to the country's Integrated Disease Surveillance Programme, which maintains data on heat waves and related mortality and morbidity, the number of states affected by heat waves increased from 9 in 2015 to 23 in 2019 (NDMA, 2019). Rise of mortality and morbidity due to heat waves can be attributed to the effects of extremely high temperature and relative humidity, and an increase in population density in urban agglomerations.

A heat wave is a period of prolonged abnormally high surface temperature. Heat waves may span several days to several weeks and are significant causes of weather-related mortality. Globally, the increasing frequency and intensity of heat waves observed since the 1950s have been associated with climate change.⁴

¹ Fifth assessment report of working Group I, 2014

² World Health Organization. (2014). Quantitative risk assessment of the effects of climate change on selected causes of death, 2030s and 2050s: World Health Organization.

³ V. K. Desai, S. Wagle, S. K. Rathi, U. Patel, and H. S. Desai, "Effect of ambient heat on all-cause mortality in the coastal city of Surat, India," *Current Science*, vol. 109, no. 9, pp. 1680–1686, 2015.

⁴ Rafferty, John P. "heat wave". Encyclopaedia Britannica, 5 Dec. 2018, <https://www.britannica.com/science/heat-wave-meteorology>. Accessed 18 February 2022.

The India Meteorology Department (IMD) declares a heat wave when a particular weather data collection station reaches 40 °C or more in the plains, 37 °C or more in coastal stations, and 30 °C or more in hilly regions. Soaring temperatures cause a large number of deaths in various parts of India. Deaths caused by heat waves were found to be higher than those resulting from avalanches, exposure to cold, cyclone, tornado, starvation due to natural calamity, earthquake, epidemic, flood, landslide, torrential rain and forest fire. From the perspective of disaster preparedness, it is important to note that deaths from heat strokes occur every year. With rising temperatures, the numbers are likely to increase⁵.

1.1 Heat stress is more severe in urban areas

The impact of extreme heat is severe in urban areas. This is because in cities heat is trapped near the earth's surface because concrete buildings, roads, and other city infrastructure absorb solar energy; rapid urbanisation causes greater degree of energy-intensive activities, and at the cost of the city's green cover. These factors affect the micro climate creating pockets of higher temperatures. These pockets are known as urban heat islands or heat hotspots.

Added to this, there is increasing population density, which makes the impact of extreme heat even more severe in urban areas. Across India, population density increased phenomenally in the past few decades as a result of rapid industrialisation and urbanisation. And in some states, like Gujarat for example, this growth has been more rapid. While Ahmedabad, Surat and Vadodara led the state's industrialisation, Rajkot emerged as a major hub of small-scale industries with more than 43,000 units located in the city.⁶ It is counted among the fastest growing cities in the world.

People have been flocking to Rajkot drawn by its booming industrial and service economy. Rajkot is the one of the largest cities in Gujarat in terms of population as well as area. Covering an area of nearly 170 sq km, the city has a population of 1.2 million as per the Census of India of 2011. This document uses the name Rajkot to refer to the city and not the district of Rajkot of which it is the headquarters.

⁵ Kumar A., Singh D. Heat stroke-related deaths in India: An analysis of natural causes of deaths, associated with the regional heatwave. J. Therm. Biol. 2020;95:102792. doi: 10.1016/j.jtherbio.2020.102792. - [DOI](#) - [PubMed](#)

⁶ Rajkot Urban Development Authority (RUDA)

Rajkot is situated in the western part of India in Saurashtra, the peninsular region of Gujarat on the Arabian Sea coast; the entire Saurashtra region is traditionally low on water. The city is characterized by a semi-arid climate. Summers here are hot and dry, and last from March to June. The temperature often crosses 42 °C and soars to 45 °C even. The city is located on the banks of Aji river; it remains dry through the year except during monsoon which lasts from July to September. Winter starts in late November and is mostly pleasant.

The fast pace of development of Rajkot city also led to rapid increase in slum pockets and squatter settlements across the city. With summers becoming more intense and accompanied by heat waves that are more frequent and of longer duration, this segment of the city population is rendered even more vulnerable as they now have to cope with the dangers of heat stress.

1.2 Impact of heat

As temperatures in Rajkot record a steady rise every summer cases related to heat stress are also increasing. Extreme heat can intensify the health effects of heat-related stress, like heat exhaustion, dehydration and heatstroke. Hot days and nights contribute to heat stress, heat-related deaths, reduced labour productivity and can exacerbate poverty. While everyone can be negatively impacted by extreme heat, certain people such as workers, especially those working outdoors in the sun or engaging in physical labour, the elderly and those with chronic illnesses, children and pregnant women are particularly at risk of suffering harm during such hot spells⁷.

What is heat stress?

Heat stress occurs when the body is unable to maintain a healthy temperature because it has been exposed to extreme heat. Common features of heat stress are exhaustion, heat rashes, fever, leading to loss of energy and reduced ability to work.

Given the scale and degree of suffering, heat waves need to be treated as a calamity, requiring a systematic approach. Heat waves require a comprehensive heat stress action plan (HSAP) that will take into account the varying degrees of hot weather within the city. For the heat stress action plan to be effective the most vulnerable sections will need to be identified. It is understood that the areas experiencing urban heat island effect will be warmer compared to

⁷ Sarah Opitz-Stapleton. September 2014, Da Nang, Vietnam: Climate Change Impacts on Heat Stress by 2050

other parts of Rajkot city. The heat stress action plan will therefore need to first identify the slums and settlements of the poor in these warmer pockets and understand the true nature of their vulnerability to heat stress.

Integrated Research and Action for Development (IRADe), a leading research institute and think tank, collaborated with the government of Gujarat, the city administration of Rajkot and the Indian Institute of Public Health, Gandhinagar (IIPH-G) to develop a climate adaptive heat stress action plan for the city. This plan is spatially differentiated and identifies the vulnerable wards lying in heat hot spots, factors leading to vulnerability and coping mechanism for municipality wards. The plan is gender sensitive because for various reasons women pay a higher price when there is extreme heat. The heat stress action plan spells out affordable adaptation strategies to help city administrators cope better with the changing climate of urban Rajkot. It also clearly demarcates the role of each government agency and other stakeholders towards better management of heat waves. The International Development Research Centre (IDRC) of the Government of Canada and Rajkot Municipal Corporation supported developing Heat Action Plan for Rajkot.

1.3 Heat waves

IPCC's report, *Climate Change 2021: the Physical Science Basis*, "shows that emissions of greenhouse gases from human activities are responsible for approximately 1.1°C of warming since 1850-1900, and finds that averaged over the next 20 years, global temperature is expected to reach or exceed 1.5°C of warming". Recommending stringent actions to limit the rise of temperature by the end of the current century to 2 °C from pre-industrial times IPCC projects a dire picture for the future. In this scenario, the frequency of heat waves in India is projected to increase 30 times the current frequency. The duration of heat waves is also expected to increase 92- to 200-fold. The trend has already begun. Summertime daily peak temperatures are higher, and heat waves are more frequent, of longer duration and more intense.

Since the 1950s there has been an increase in the frequency and intensity of heat waves globally. This had a severe impact on the densely populated cities of India, where a large number of people live on low incomes in slums characterized by congested and poorly ventilated living spaces. The increasing severity of heat waves extracts a heavy price as it

makes India's urban poor vulnerable to heat. According to a study published in Lancet in 2021, India had the highest burden of mortality associated with high temperatures between 2000 and 2019.⁸ Between 1992 and 2015, heat waves caused 24,223 deaths across the country.⁹

Even when it did not kill, the heat stress affected the health, productivity and livelihoods of vulnerable populations. Despite the heavy toll, the government gave focused attention to heat waves only in 2010, triggered by the historic heat wave in Ahmedabad city in Gujarat. Extreme heat is a silent disaster Heat Stress Action Plan

The 2010 devastating heat wave of Ahmedabad was a wake-up call. The city temperature rose to 46.8 °C (116 °F) and caused a large number of deaths in one week alone. Experts estimated the heat contributed to more than 1,000 deaths¹⁰. Preliminary analyses suggested that an excess all-cause mortality of at least 30 per cent was associated with this event¹¹. The high toll created the urgency to devise a strategy, and led Ahmedabad Municipal Corporation to launch South Asia's first heat action plan in 2013.

In 2016 India formulated a National Disaster Management Plan for the first time. It heeded heat waves as a serious disaster and recommended vulnerable cities be made heat friendly. Recognizing the growing threats from rising temperature, cities across India adopted heat action plans the same year.

2016 national disaster management plan laid down India's vision for reduction of disaster risk and losses in lives, livelihoods and health. It created a roadmap for the government and communities to work in tandem to adapt to risks in a systematic and proactive manner. This was also in line with the objectives of the Sendai Framework for Disaster Risk Reduction 2015-

⁸ [https://doi.org/10.1016/S2542-5196\(21\)00081-4](https://doi.org/10.1016/S2542-5196(21)00081-4)

⁹ <https://ndma.gov.in/sites/default/files/IEC/Booklets/HeatWave%20A5%20BOOK%20Final.pdf>

¹⁰ Heat-Related Mortality in India: Excess All-Cause Mortality Associated with the 2010 Ahmedabad Heat Wave. PLoS ONE 9(9): e109457. <https://doi.org/10.1371/journal.pone.0109457>

¹¹ Knowlton, K., Kulkarni, S. P., Azhar, G. S., Mavalankar, D., Jaiswal, A., Connolly, M., Nori-Sarma, A., Rajiva, A., Dutta, P., Deol, B., Sanchez, L., Khosla, R., Webster, P. J., Toma, V. E., Sheffield, P., Hess, J. J., & Ahmedabad Heat and Climate Study Group (2014). Development and implementation of South Asia's first heat-health action plan in Ahmedabad (Gujarat, India). *International journal of environmental research and public health*, 11(4), 3473–3492. <https://doi.org/10.3390/ijerph110403473>

2030, a UN agreement which 187 countries including India signed in March 2015 in Sendai, Japan.

Since inception the focus of heat stress action plans has been heat vulnerable populations. City municipal corporations collaborated with various development partners to deliver on the following:

- build awareness on health risks related to extreme heat
- develop a heat early warning system that alerts communities
- targeted interventions for vulnerable groups

A 2018 evaluation of Ahmedabad's heat action plan showed that the city has been able to prevent more than 1,100 heat related deaths every summer.¹² The key to this achievement is identifying where the vulnerable populations live. This is becoming more and more challenging as expanding city limits leads to a rise in the number of slums and squatter settlements, which are spread across various locations. Not just that, as temperatures grow extreme and cities reel under urban heat island effect there are different temperatures in different pockets of a city. Some pockets are warmer than other parts. It is critical therefore to identify the right locations – the pockets where the poor live and are warmer than other parts of the city – in order to prepare a climate adaptive heat action plan.

1.4 Climate Adaptive Heat Action Plan

Rajkot Heat Action is first ward level climate adaptive plan in South Asia. The revised heat action plan is climate adaptive – which considers climatological variance and future climate scenarios. It considers makes sure the targeted intervention is a right fit and designed for the heat vulnerable population of a city. It first identifies the heat hotspots of the city, locates the poor people in these pockets, and assesses the nature and status of their vulnerability to extreme heat. In the first phase, a climate adaptive heat stress action plan carries out a comprehensive climate profiling of the city, surveys the socio-economic status including nature of work of the people living in the heat hotspots, and then identifies the most vulnerable municipality wards of the city.

¹² Building Resilience to Climate Change: Pilot Evaluation of the Impact of India's First Heat Action Plan on All-Cause Mortality, 2018 <https://doi.org/10.1155/2018/7973519>

In the second phase, the plan draws up a framework for implementation, coordination and evaluation of extreme heat response. It clearly demarcates the role and responsibility of each government agency, including the ward level administrative unit of the city municipality, or the urban local body. It also spells out the role of other stakeholders like community-based organizations, NGOs and the community in ensuring better preparedness, mitigation and management of heat stress.

A climate adaptive plan adds the following advantages to a city's efforts to mitigate heat stress:

- It provides a climate-sensitive framework for implementation, coordination and evaluation
- Alerts populations at risk of heat-related illness in places where extreme heat conditions prevail, by adopting dynamic climate intelligence

2. How heat waves are affecting Rajkot

2.1 Heat Waves in Rajkot city

The frequency of heat waves in the city increased substantially during 2001-10, as compared to earlier decades (Ray, Chincholikar, & Mohanty, 2013).

The trend of heatwave occurrence in the city during the past decades indicates a gradual increase in the heatwave days and severe heat days from 2 to 13 days (March – June). It is noteworthy that of the reference period from 2001 to 2021, heatwave was witnessed in almost every year. In comparison, the earlier years showed an intermittent incidence of heatwaves of relatively shorter durations.

Over the years we see the increase in the frequency and the number of heatwave days. There was a 49% increase in heatwaves days between 1971 to 2021.

Decadal cumulative frequency of heatwaves days has oscillated from 6 days (1970-80) to 22 days (1990-2000) (Ray et al, 2013¹³) and again 39 days during 2001-10 to 66 days during 2011-21 (IMD) constituting a percentage increase of 51% in the number of heatwaves.

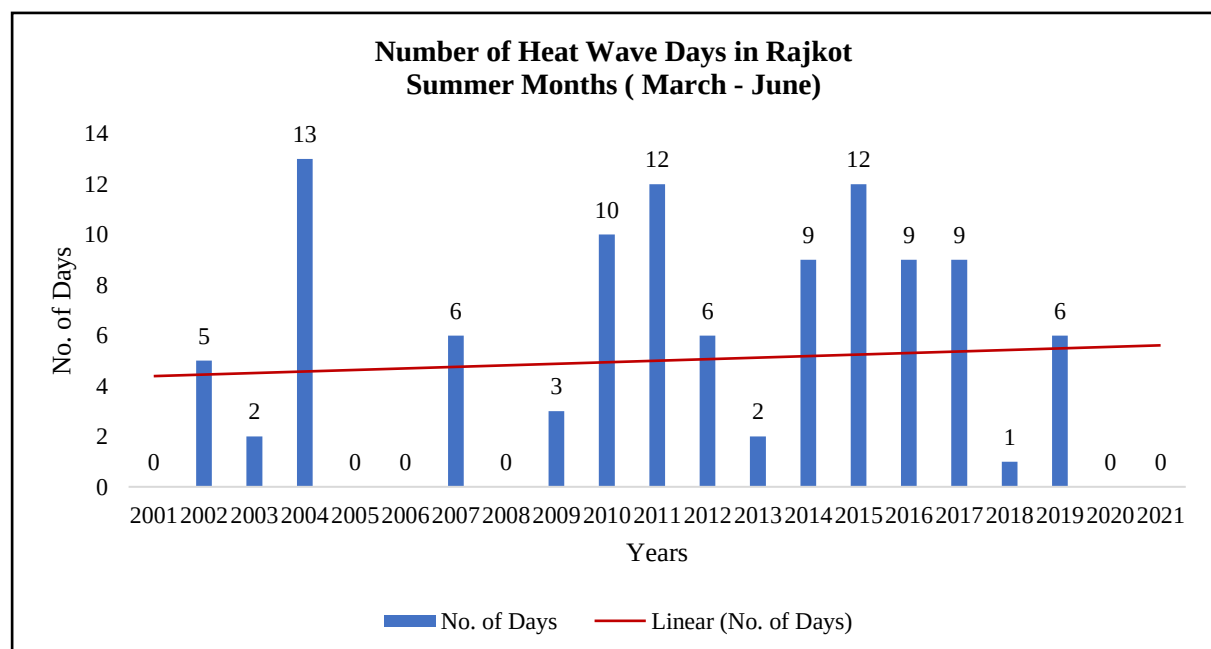


Figure: Trend in Heat Wave Days, (2001-21) Rajkot
Source: IMD

¹³https://www.researchgate.net/publication/278021314_Analysis_Of_Extreme_High_Temperature_Conditions_Over_Gujarat

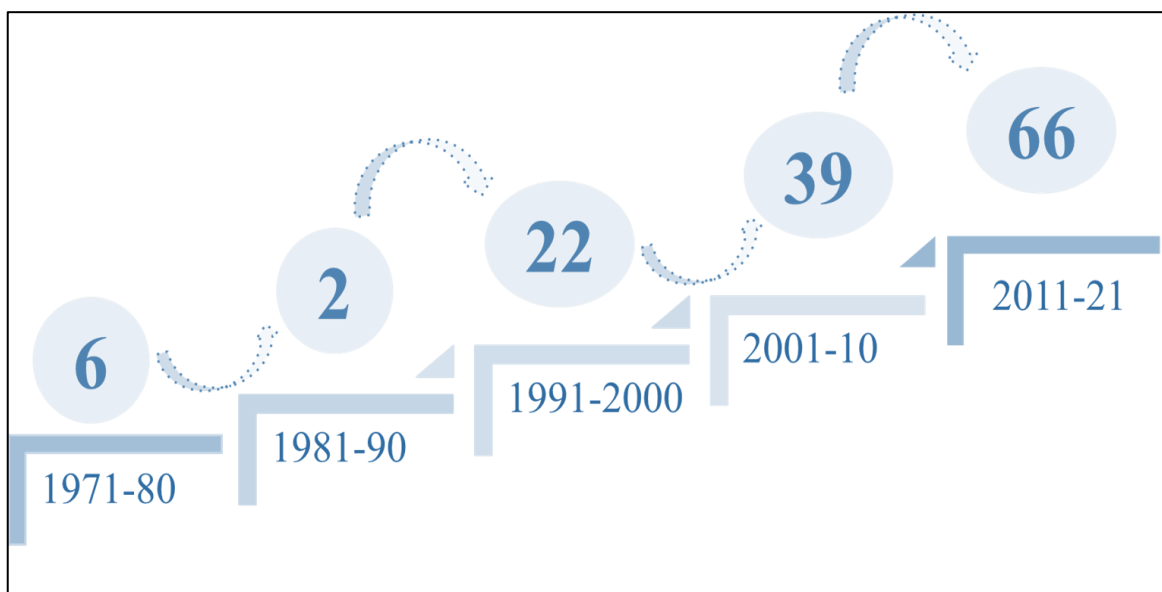


Figure: Decadal cumulative frequency of heatwaves days (1981-2021)

Source: Ray et al & IMD

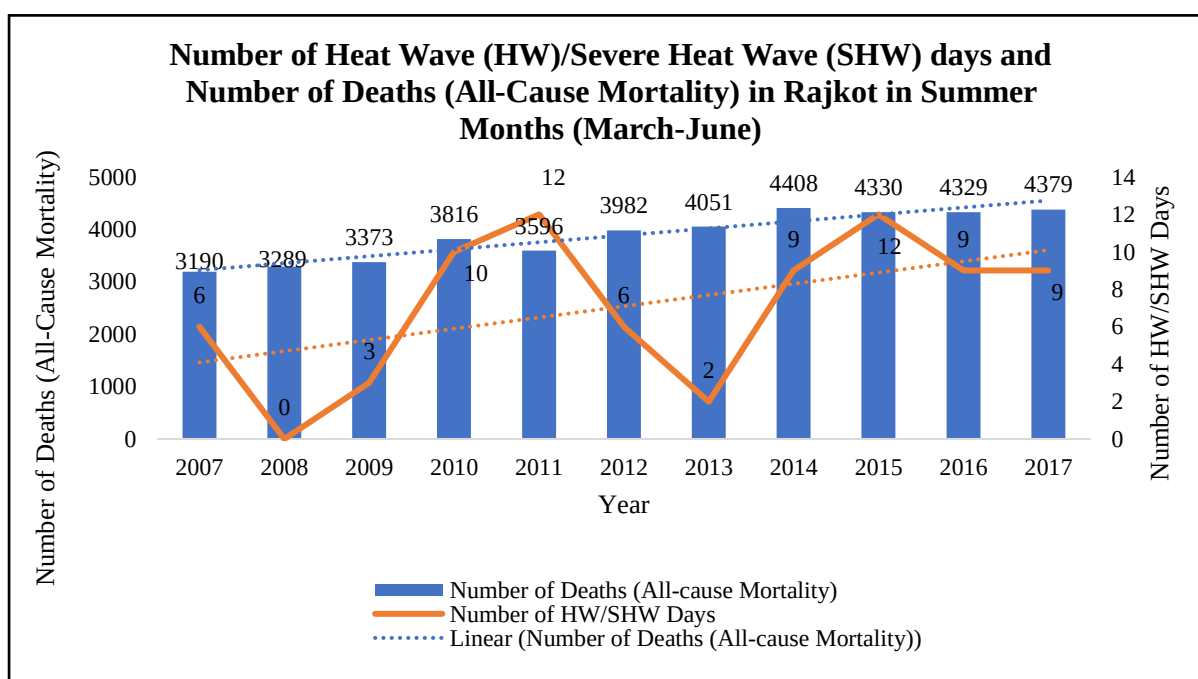


Figure: Co-relation between Heat Wave Days and Mortality Rate (2007- 2017)

Source: IMD & Heath Dept. Rajkot Municipal Corporation

All-cause mortality cases during the summer months have shown a rise with the rising summer temperature in the past few years. The city's Disease Surveillance System and Death Record System does not record deaths due to heat wave or heat stress. However, due to the rise in

temperature in the last few years in the city of Rajkot the cases related to heat stress have been increasing during the summer months.

The risks to human lives as a result of heat stress grew significantly in Rajkot in recent years as it expanded in area and population. From 23 municipality wards in 2015 the city now has 18 wards, with an increase in the increase in the slum pockets and scattered settlements across the city from **118 slums in 2012 (SFPoA, Rajkot, 2012)** to **145 slums as of 2017 (Gujarat Government Gazette, 2017)**.

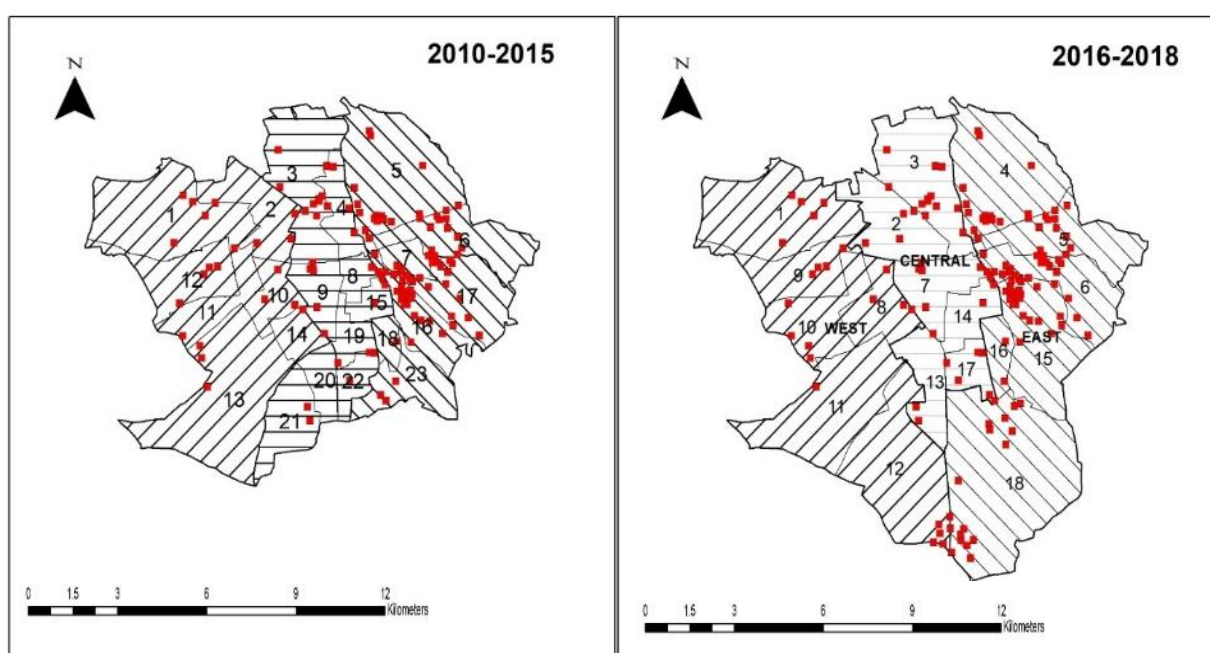


Figure: Slum Pockets (2010-15 & 2016-18), IRADe

2.2 City Climatological Variance

Rajkot has a semi-arid climate, with hot, dry summers from mid-March to mid-June. A climatological analysis clearly showed significant increase in the city's temperature and moderate decline in humidity levels. This indicates that the city is getting drier and warmer.

The climatological parameters that influence heat waves are high temperatures and relative humidity of a region. The parameters IRADe analysed were maximum temperature, minimum temperature and relative humidity. The climate parameters show a sharp increase in the month

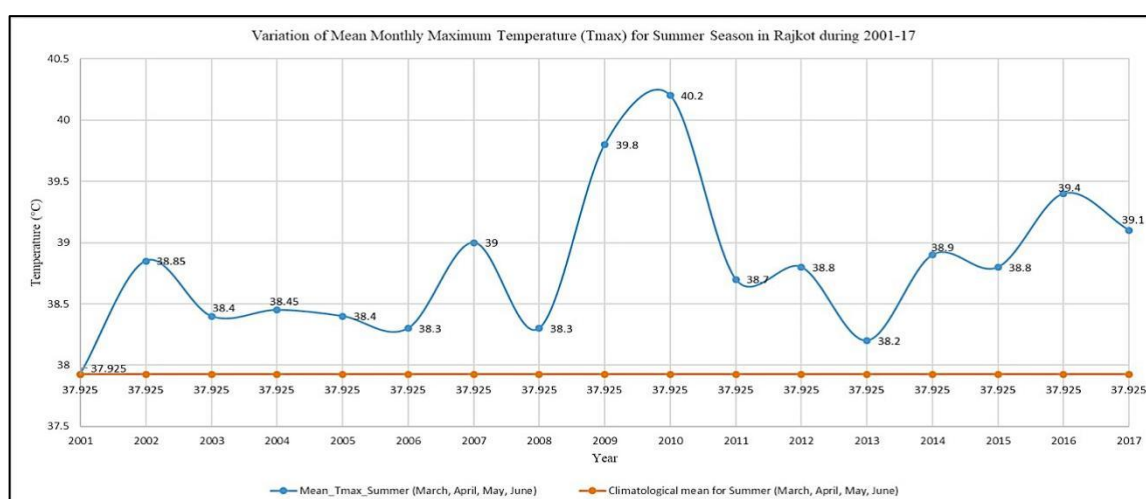
of March, which suggests Rajkot is experiencing relatively more heat in March. Increase in minimum temperature along with humidity will lead to increase in heat stress.

In order to understand the changing weather patterns of Rajkot, IRADe conducted a study of the city's summer climatology (March, April, May and June). It studied IMD data from 2001 to 2017 pertaining to temperature and relative humidity. IRADe analysed the city's temperature data to determine trends in maximum and minimum temperatures over 122 days in March, April, May, and June for the 17-year period between 2001 and 2017, and compared it against the city's long-term climatological mean (1905 to 2000) for the corresponding months to determine the deviation in the city's temperature trends.

Variation of Maximum and Minimum Temperatures for summer season

The average increase in Tmax value for entire summer period over the study duration of 17 years is 1.095°C, in Rajkot. The average increase in Tmin value for the entire summer season over the 17-year period comes out to be 1.43 °C, with maximum deviation in seasonal Tmin value was observed in March (1.7 °C).

Month (2001- 2017)	Tmax (° C)	Deviation from Mean Tmax (° C)	Avg Increase in Mean Tmax (° C)	Tmin (° C)	Deviation from Mean Tmin (° C)	Avg Increase in Mean Tmin (° C)
March	35.2	1.09	1.095	18	1.7	1.43
April	38.6	1.2		21.7	1.6	
May	40.3	0.84		24.8	1.2	
June	37.6	0.36		26.1	0.7	
Tmax- Maximum Temperature, Tmin – Minimum Temperature, Source: National Data Centre. IMD						



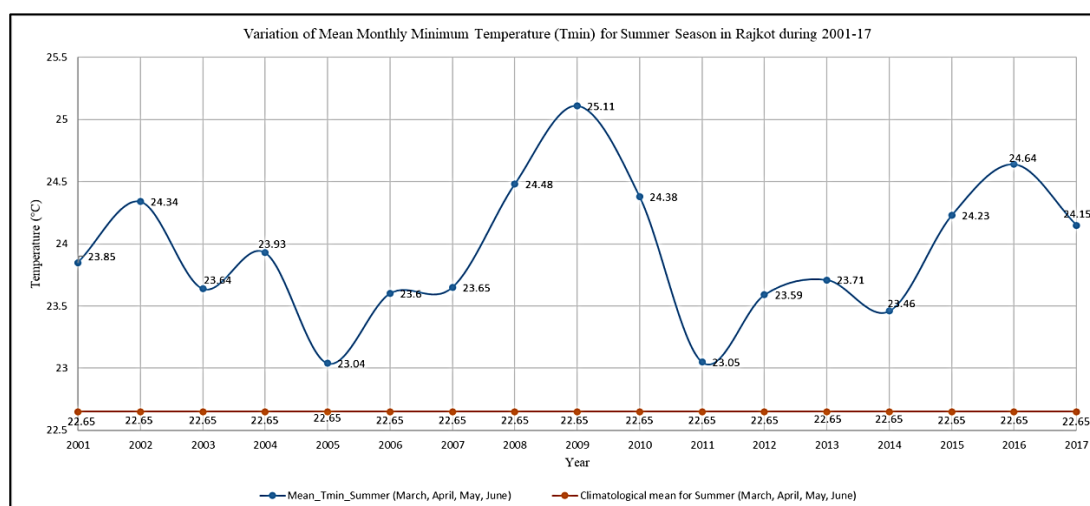


Fig: Variation of Tmax and Tmin for summer season in Rajkot from 2001 to 2017

Similarly, the study analysed trends in relative humidity in the morning (08:30) and evening (17:30) over March, April, May, and June from 2004 to 2017 against the long-term climatological mean relative humidity.

Variation of Mean Monthly Relative Humidity (RH) measured in the morning at 8:30 AM for Individual Summer Months

Maximum deviation for RH (830) was observed in March (+1.33), followed by April (+1.12). The average mean RH (830) over the 17-year study period comes out to be +73.705. Maximum deviation in RH (1730) was observed in June (+2.46). The average mean RH (830) over the 17-year study period was +30.66.

Month (2001- 2017)	RH (830)	Average Mean RH (830)	Deviation from Mean RH (830)	RH (1730)	Average Mean RH (1730)	Deviation from Mean RH (1730)
March	+69	+73.705	+1.33	+21.33	+30.66	-1.27
April	+71.33		+1.12	+21		-0.48
May	+75.16		+0.42	+30.16		-1.40
June	+79.33		+0.29	+50.16		+2.46
RH – Relative Humidity. Source: IMD						

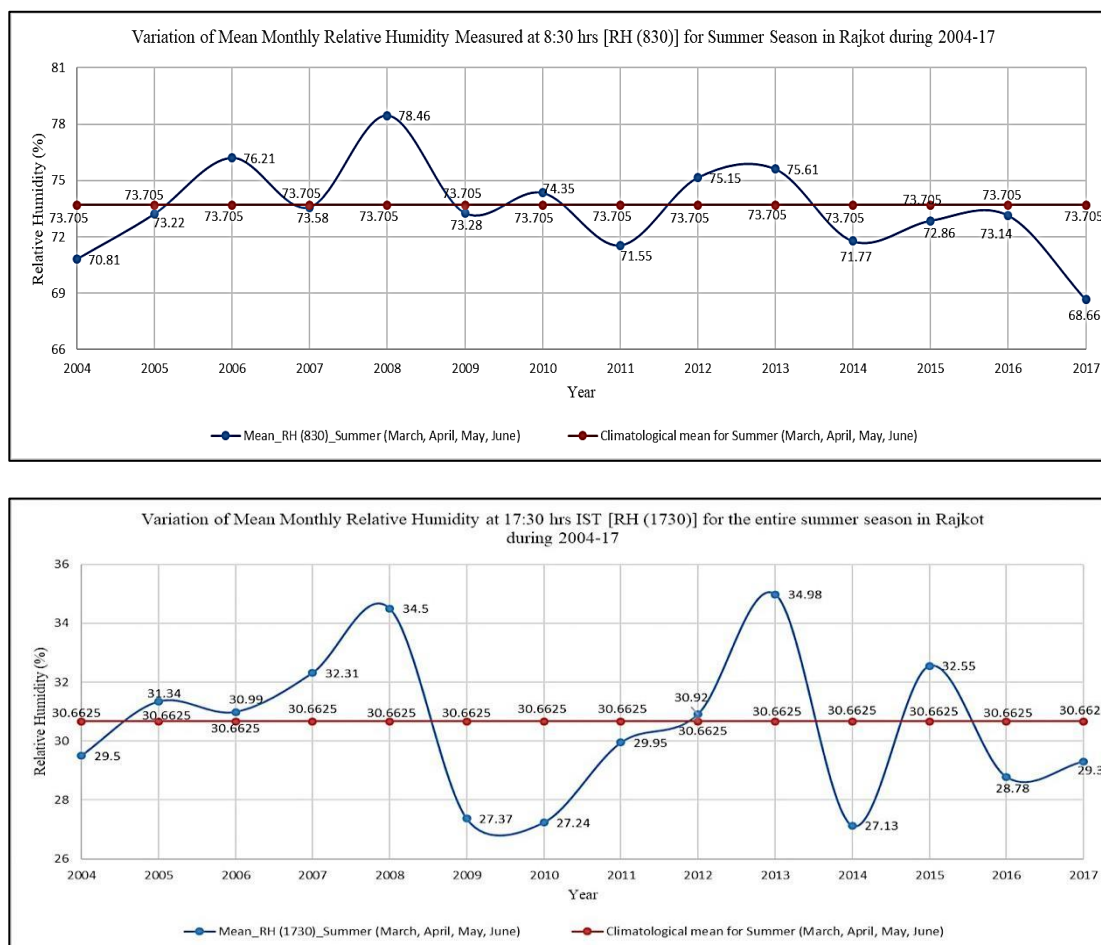


Fig: Variation in RH at 8:30 hrs for entire summer period in Rajkot from 2004-2017

The climate parameters show a sharp increase in the month of March, which suggests Rajkot is experiencing relatively more heat in the month of March. Increase in minimum temperature along with humidity will lead to increase in Heat Stress.

Further, the study determined the variation in these climate parameters for all 122 days of summer together for the study duration. The analysis of climatological parameters presented a trend that indicates hotter and drier climate in Rajkot. It is more pronounced for March and April with significant rise in mean maximum and minimum temperatures, and decline in relative humidity.

There is clear evidence of climate variability leading to increase in number of heat wave events as well as early arrival of hot days. Local authorities need to be prepared earlier in the month of March. March is usually considered the transition month between winter and summer. A sharp increase in temperature and lower relative humidity in March will not provide people sufficient time to acclimatize. This in turn may lead to an increase in public health emergency for heat stress management.

3 Vulnerability assessment

Several slums and low-income squatter settlements in Rajkot are located in areas that experience temperatures higher than the city average. In order to prepare a climate adaptive heat stress action plan for the city it was necessary to first identify its heat hotspots. Mapping a city's heat hotspots helps to direct interventions where they are most needed during heat waves.

3.1 Heat hotspots of Rajkot city

Urban Heat Island (UHI) is a phenomenon where higher temperatures are observed in the city centres as compared to its surrounding areas. The UHI effect has emerged as a potential hazard for rapidly urbanizing countries like India. It adversely affects energy usage, public health, economy and also contributes to climate change (Tyagi et al, 2022)

Where is the hottest area in city?

1. **Built Environment:** The built environment within a city absorbs heat and radiates it out due to the use of materials such as concrete, asphalt, and steel. This phenomenon is most commonly known as urban heat island effects in city.
2. **Density of structure:** It is observed that the factors of density and variety of structures, as well as the lack of natural space, the informal settlements are consistently hotter than the surrounding city. As a result, residents in these places are more likely to be exposed to intense heat and may be at risk of heat-related morbidity and mortality.
3. **Less green cover:** The urban areas witness warmer temperatures during summer nights, compared to the interior parts of India. The nights are warmer in urban areas owing to the absence of evapotranspiration and thermal storage in concrete structures. As a general estimation, areas with more green spaces tend to be cooler.

When working with local governments to develop interventions, it's crucial to think about how temperatures might change inside a city, where they're higher, and where the most vulnerable individuals are. These variables not only help in developing targeted interventions but also helps in issuing timely mitigation measures to minimise heat wave impacts.

Partnering with stakeholders can be a helpful way to identify specific information about the city.

In order to assess the spatial distribution of heat stress at the ward level in Rajkot, IRADe worked in collaboration with India Meteorological Department (IMD) and Rajkot Municipal Corporation (RMC). It first mapped thermal hotspots through remote sensing using land surface temperature (LST) images procured from Landsat 8, an American earth observation satellite. It then validated the LST derived from satellite data with ambient air temperature recorded by the IMD station in Rajkot as well as the data received from 20 Automatic Weather Stations (AWS) installed within the city by the municipal corporation. IRADe then superimposed these surface temperature maps on the ward-boundaries map and marked the wards with temperature above 42 °C. These hotspots were identified based on the surface temperature recordings in May and June of 2017 and 2018. Rajkot Municipal Corporation provided data related to municipal wards and slum locations.

The hotspot maps so generated are useful for policymakers and city administrators in analysing the local factors contributing to heat-stress in different wards and devising mitigation options to reduce heat stress in these areas.

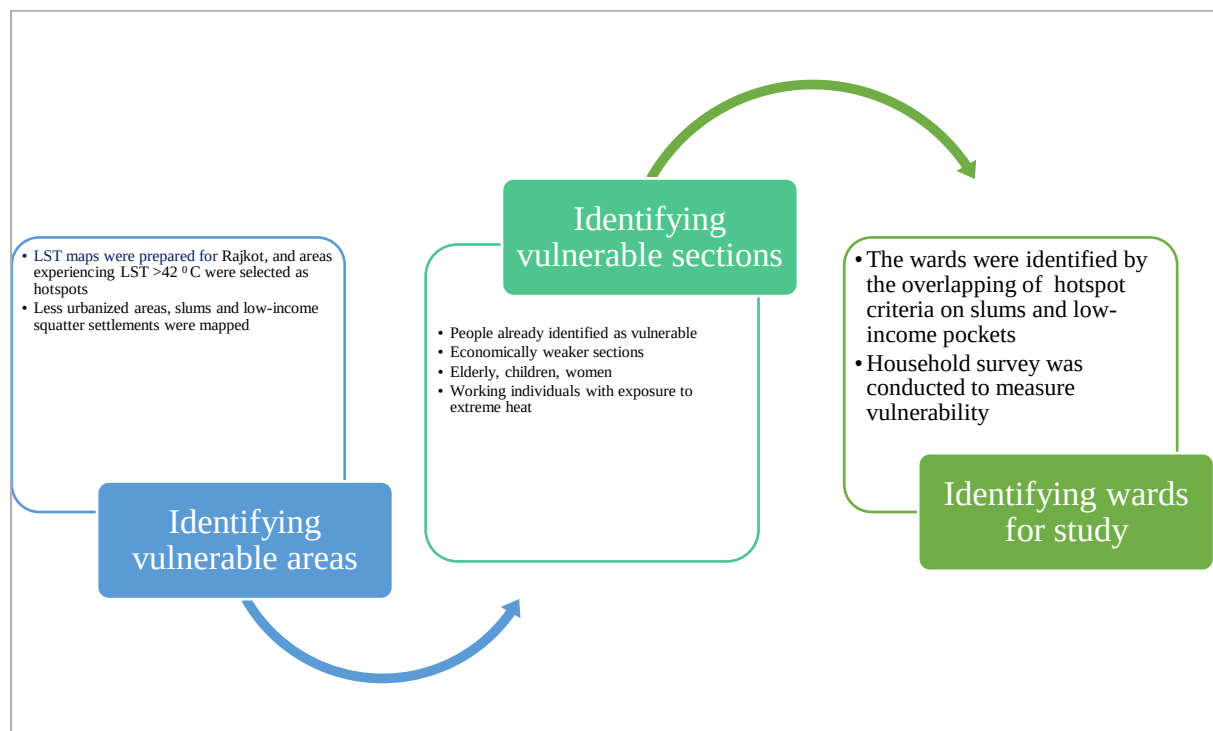


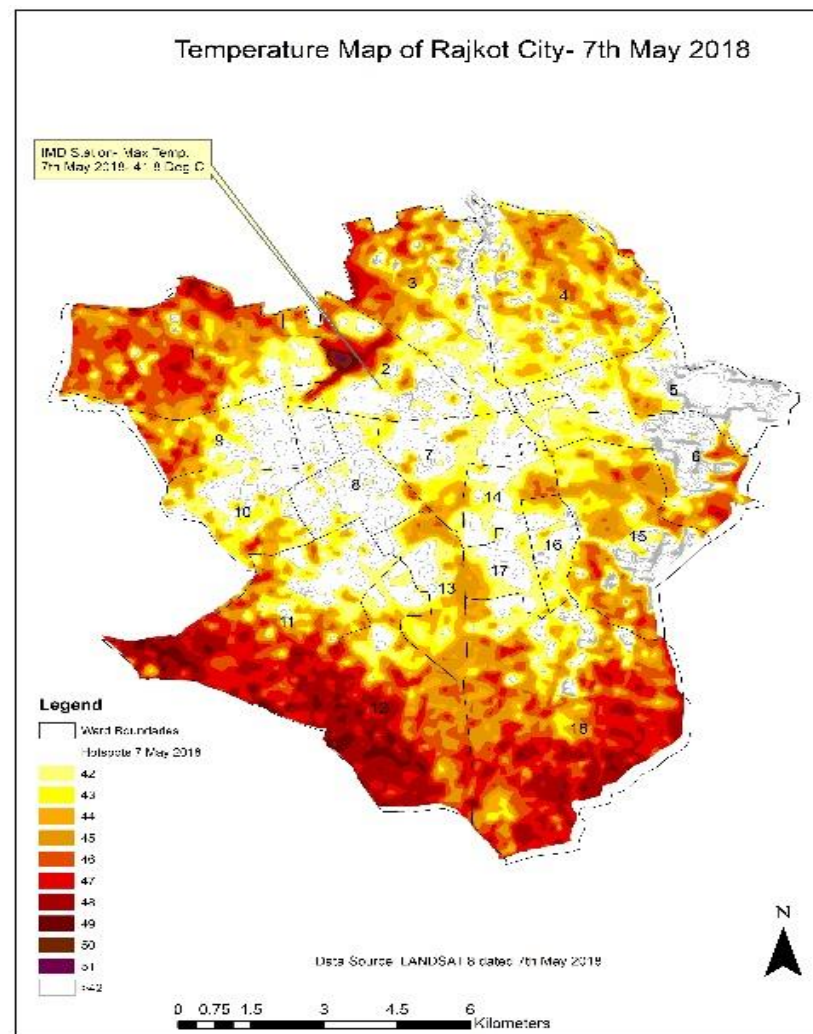
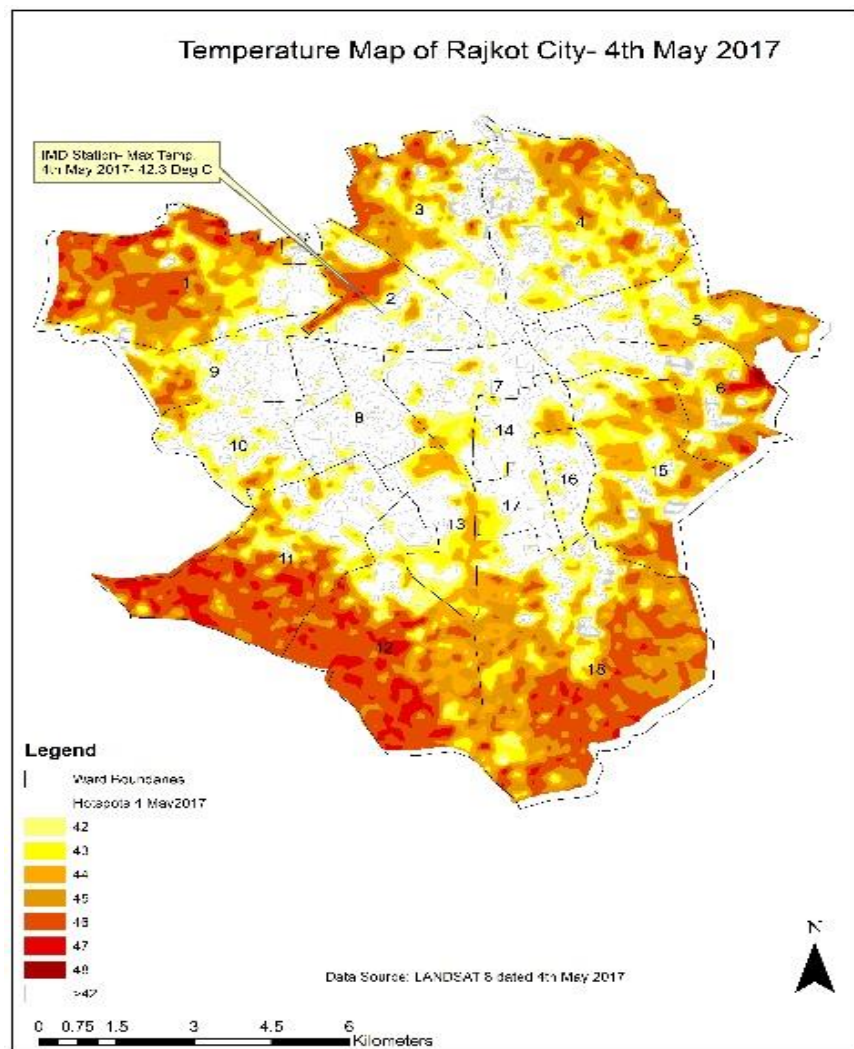
Figure: Steps to identify municipality wards for study

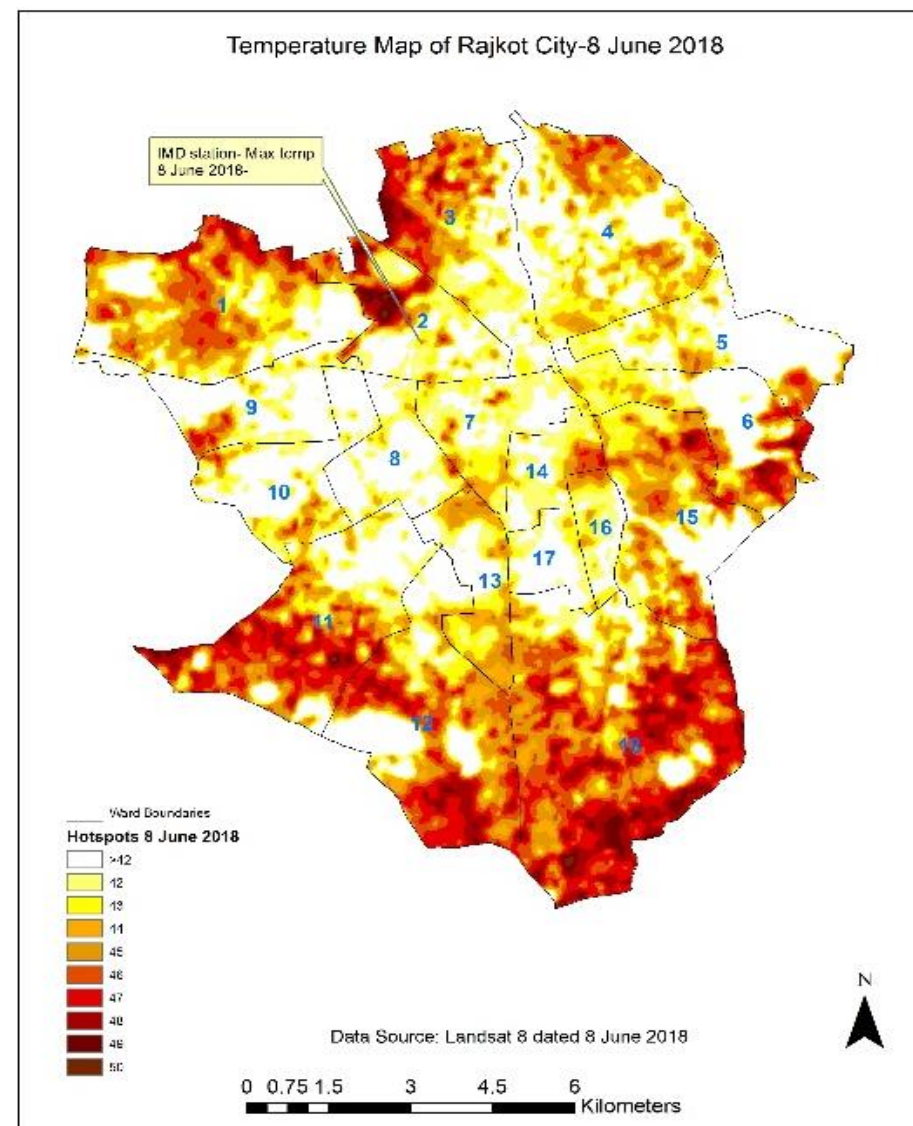
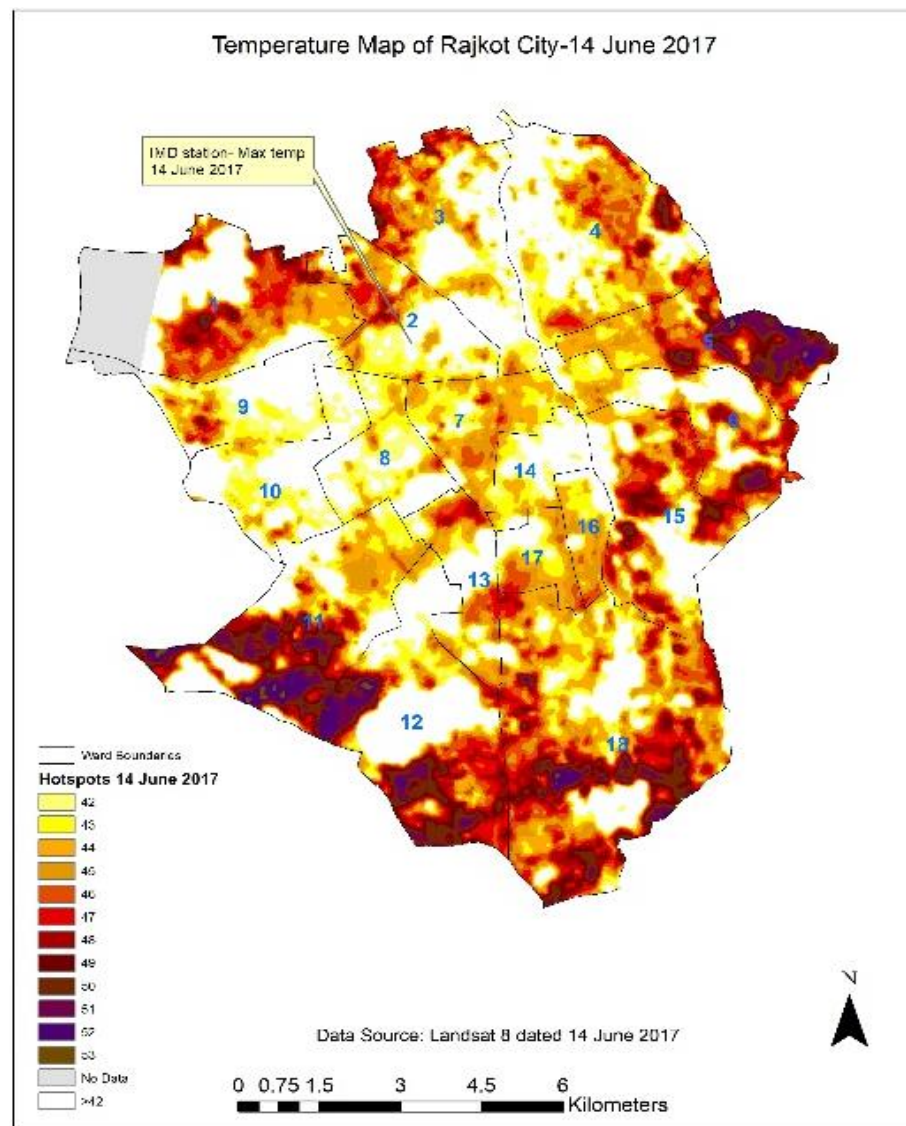
The mapping exercise marked 9 wards as heat hotspots in Rajkot city

Table: Wards identified as heat hotspots

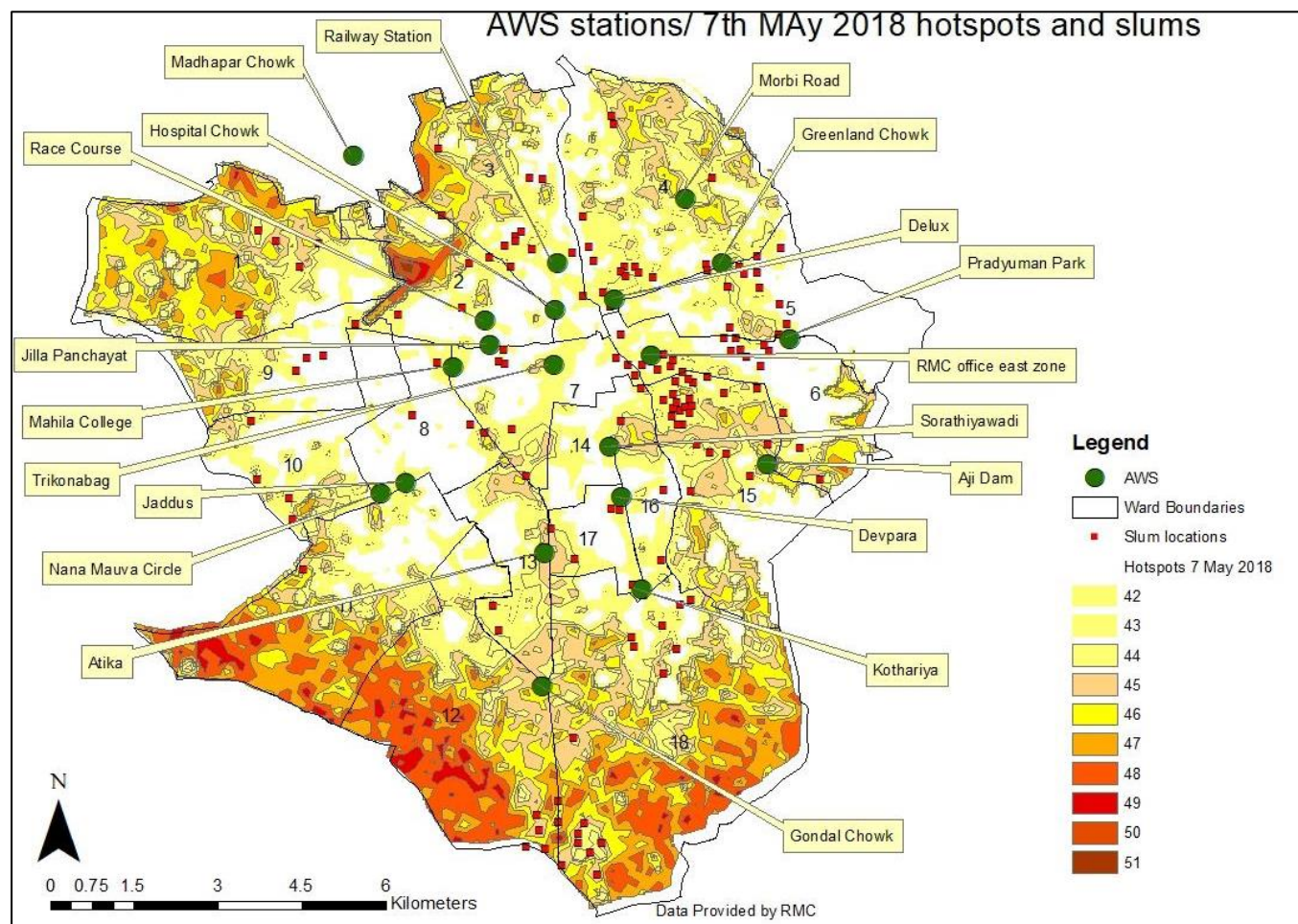
Ward name	Ward number
Rajyadhar	1
Bhimrav Nagar, Pradyuman Park, Sadhu Vasvani Road	5
Jay Bhimnagar	10
Bharatnagar	11
Bajrang Society Rashulpura	12
Ambedkar Nagar	13
Liludiwonkdi	14
BharatNagar 1	15
Shitaldhara	18

Map: Land surface temperature maps showing recordings in May and June of 2017 and 2018



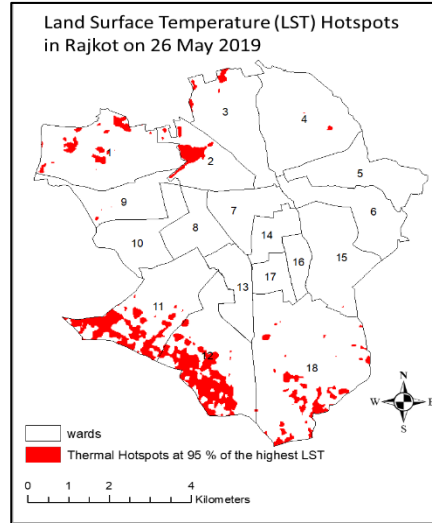
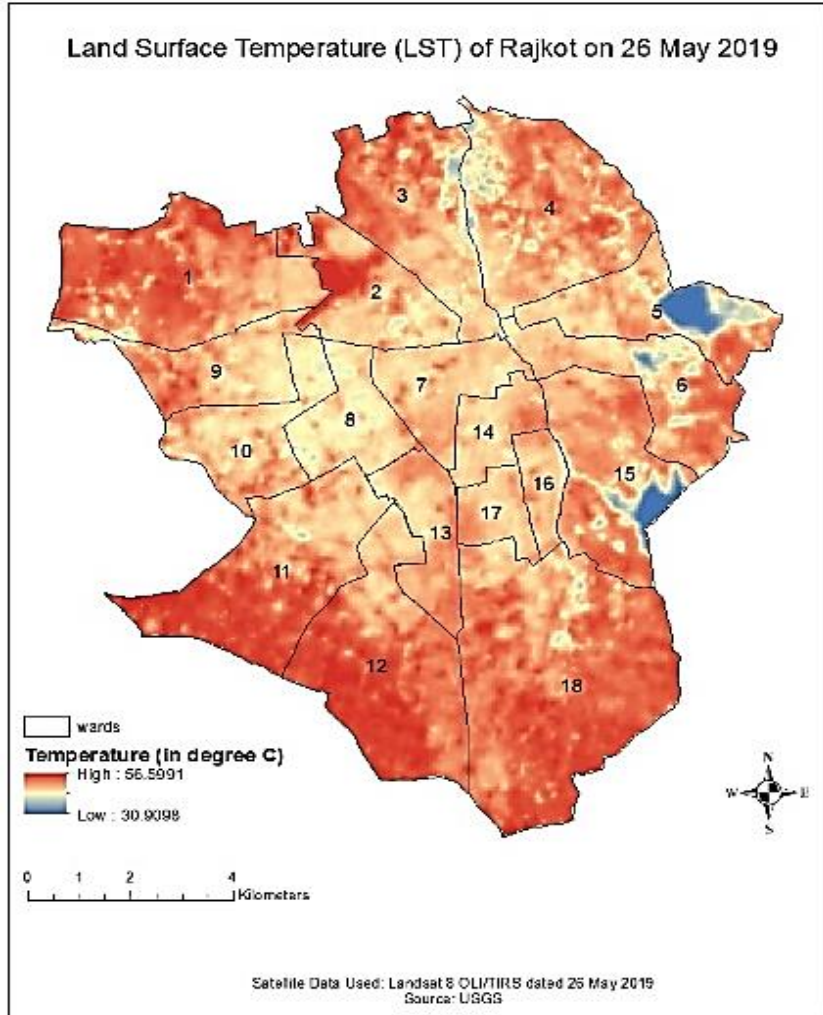


Map: Ambient air temperature across Rajkot in May 2018 recorded by weather stations



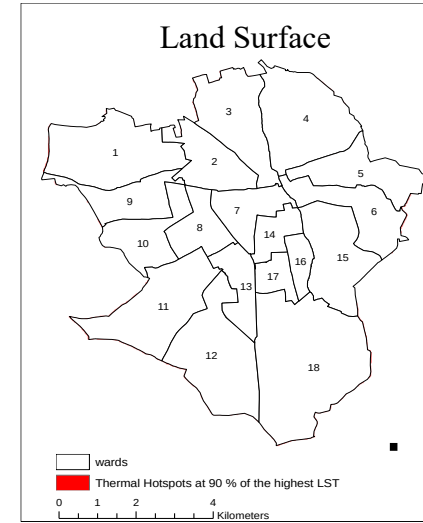
Wards with temperature above 42 degrees Celsius were delineated across the city. higher temperature is consistently experienced in ward numbers 1, 2, 3, 4, 5, 6, 9, 10, 11, 12, 15 and 18. Peripheral regions show higher temperatures than the central regions.

3.1.1 Hotspot Delineation - spatial distribution of thermal hotspots



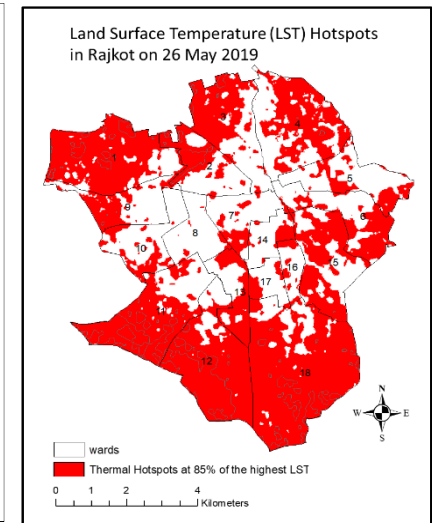
Thermal hotspot map of Rajkot showing wards with LST >53.77 °C

Wards 11 and 12 have areas falling in Category II, with roughly 26 and 34 % areas with LST in excess of 53.77 °C



Thermal hotspot map of Rajkot showing wards with LST >50.94 °C

Almost 16 wards indicate thermal hotspots at 90% of the highest LST range. Ward number 12 has nearly 68% area covered as hotspot

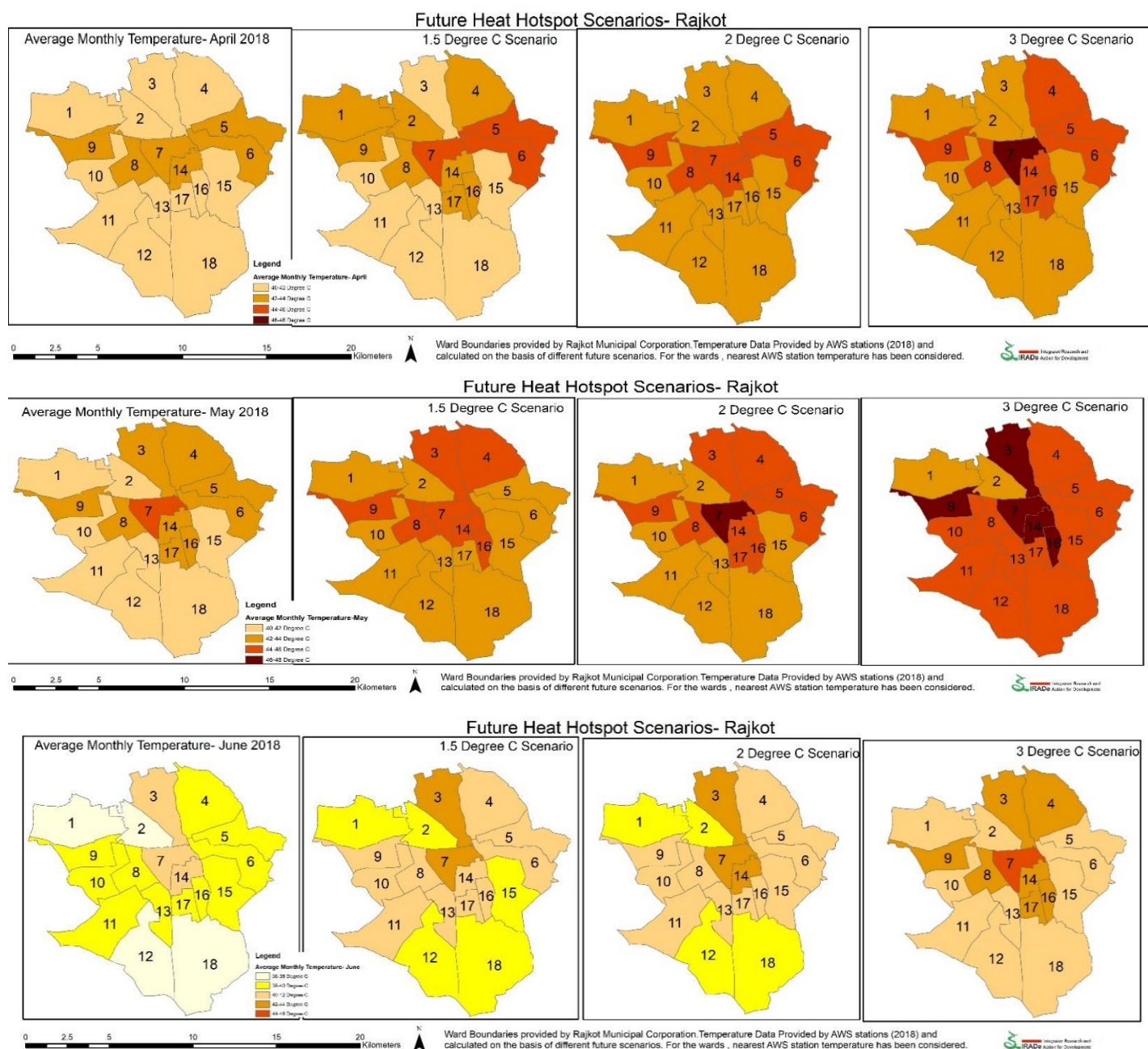


Thermal hotspot map of Rajkot showing wards with LST >48.11 °C

Except, 7 and 8 all the wards have hotspot areas in Category I and II. Wards 1, 2, 11, 12 and 18 show areas with LST more than all the percentile ranges (95-85%)

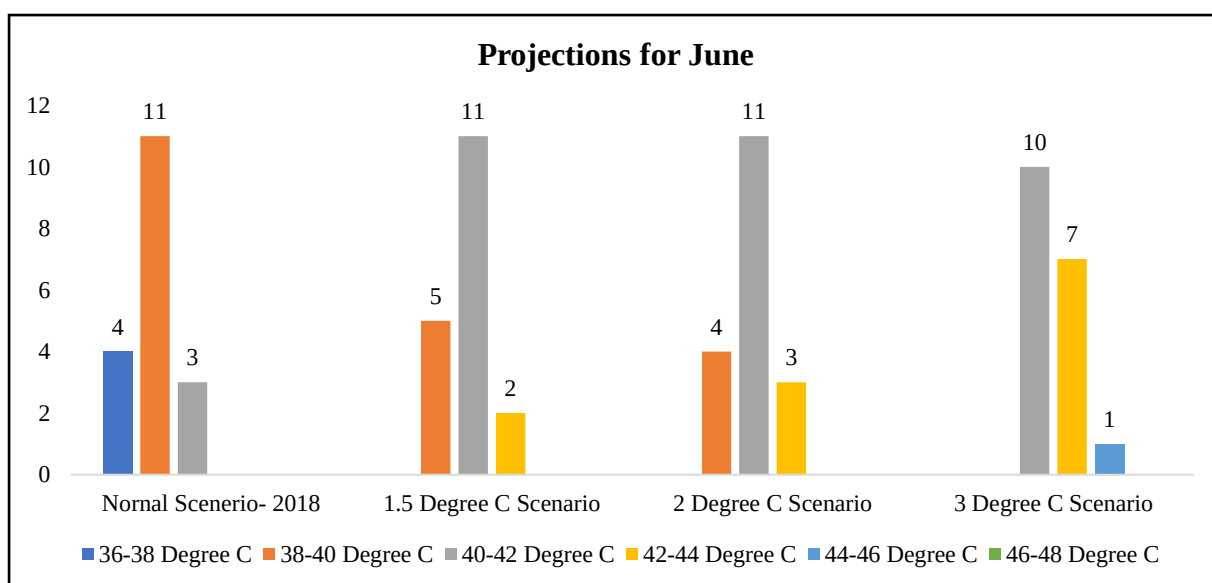
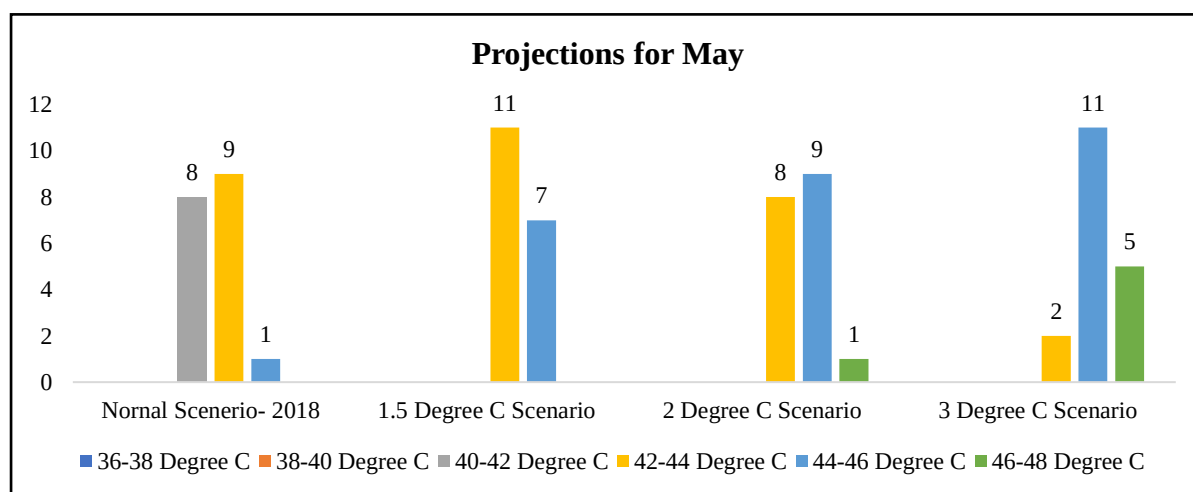
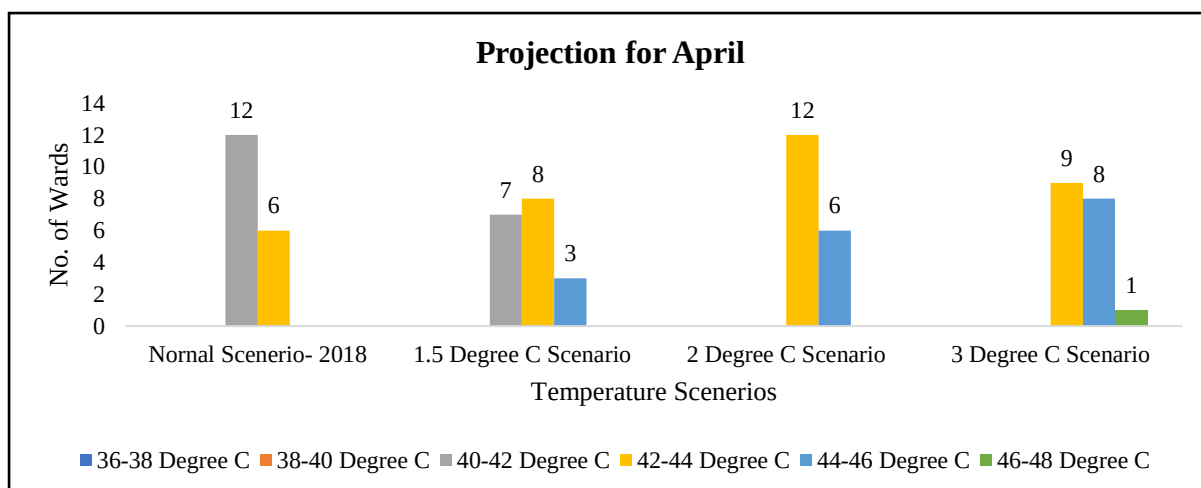
Extraction of hotspots based on percentile of highest temperature observed in the LST map. For each LST map, areas experiencing LST more than 95%, 90%, 85% of the highest LST range were extracted.

3.1.2 Future Heat Hotspot Scenarios for Rajkot- (April. May. June)



Based on the temperature data of AWS stations across the city of Rajkot (2018), the future scenario temperature rises at 1.5°C, 2°C and 3°C for the months of April, May, and June is mapped

The analysis clearly indicates that with each scenario, the number of wards in the higher temperature range is increasing and those which were earlier in the lower temperature range are moving towards higher temperature ranges.



The predictions for the future thermal hot spots across Rajkot, indicate that for the month of May Eight wards are in the temperature range 40-42⁰C, 9 wards in 42-44⁰C and 1 ward in 44-46⁰C in the Average Monthly baseline temperature of May (2018). However, in the scenario with 1.5⁰C increase in temperature, 11 Wards are in the temperature range 42-44⁰C, 7 wards are in 44-46⁰C. In the scenario with 2⁰C increase in temperature, 8 Wards are in the temperature range 42-44⁰C, 9 wards are in 44-46⁰C and 1 ward in the 46-48⁰C. In the scenario with 3⁰C increase in temperature, 2 Wards are in the temperature range 42-44⁰C, 11 wards are in 44-46⁰C and 5 wards in the 46-48⁰C.

This clearly indicates that with each scenario, the number of wards in the higher temperature range are increasing and those which were earlier in the lower temperature range are moving towards higher temperature ranges.

3.2 Mapping vulnerability

In 2001 IPCC defined vulnerability to heat as a function of: the degree of exposure to the heat hazard, sensitivity to changes in weather/climate (the degree to which a person or system will respond to a given change in climate, including beneficial and harmful effects), and adaptive capacity (the degree to which adjustments in practices, processes, or structures can moderate or offset the potential for damage or take advantage of opportunities created by a given change in climate).

After identifying the nine heat hotspots, IRADe therefore assessed the vulnerability of the wards located therein as defined by IPCC. It conducted one primary and two socio-economic surveys. The primary survey included 286 households; the two socio-economic surveys were conducted 14 days apart and included 282 and 202 households in the first and second fortnight, respectively. These two surveys served as a baseline and end line and were the basis for calculating the impact of extreme heat on health, livelihood and productivity of the vulnerable community in Rajkot.

The surveyed households included construction workers, street vendors, domestic helps and road sweepers. Each household comprised four members on average. Making door-to-door visits the survey captured information on the resident population's occupation, working hours, occupational pattern, mode of transportation, the impact of heat stress, and the coping capacity

of households. The survey used a comprehensive index of factors to assess the vulnerability of households. The following are the eight sectors and corresponding sub-sectors in the index:

S. No.	Sectors	Sub-sectors
1	Housing	Ownership
		Number of rooms
		Housing structure, and material used for floor, walls, roof
		Number of windows, and exterior wall colour
2	Cooking	Type of fuel
		Cooking area
3	Sanitation	Access to toilet
4	Water	Water source
		Water supply (in litres)
5	Electricity	Electricity supply and frequency of outages during summer
		Electricity bill and electrical appliances used
6	Health	Mapping high heat period
		Time of heat discomfort
		Heat stress symptoms
		Access to health infrastructure
		Distance from nearest facility
		Age-wise symptoms
		Health insurance
		Source of health insurance
		Reasons for not using government health facilities
		Differential impact of heat on women and men
7	Awareness	Awareness of the term 'heat stress'
		Awareness of medical facilities treating heat stress
		Awareness of mitigation strategies adopted by ULBs
8	Heat stress symptoms	Heat exhaustion, heat cramps

3.3 Survey findings

3.3.1 Housing

Poor quality housing in terms of design and materials used, common in settlements where the urban poor live, increases discomfort and health risks during extreme heat. It increases the indoor temperature as houses heat up quickly and cool down very slowly. Staying indoors during heat waves may therefore lead to other health risks in heat vulnerable housing; overcrowding too increases these risks.

Of the nine wards surveyed, seven were found highly vulnerable — these are ward numbers 1, 10, 12, 13, 14, 15 and 18. Ventilation was almost absent in these wards. Two wards — numbers 5 and 11 — showed medium range of vulnerability. As a well-ventilated house provides cooling through aeration and heat dissipation, RMC housing design byelaws need to make ventilation mandatory as an adaptive and mitigation measure.

3.3.2 Cooking

More than 90% of the adult women in the survey do the cooking for the family; their vulnerability increases when the cooking area is tiny and has no ventilation. Most of the wards surveyed tend towards high vulnerability – wards 1, 10, 14, 15 and 18 are highly vulnerable due to use of wood as cooking fuel inside the house. Wards 5, 11, 12 and 13 showed medium range vulnerability with cleaner cooking fuel.

3.3.3 Sanitation

Poor access to sanitation was more prevalent in ward numbers 14 and 15 indicating high vulnerability; this was due to lack of toilet facility in the premise. Ward numbers 1, 10, 12 and 13 showed medium vulnerability and ward numbers 5, 11 and 18 showed low vulnerability.

Improving access to sanitation and piped water supply for all households is major step for checking peak summer disease caseloads.

3.3.4 Water

All the nine hotspots in the city — ward numbers 1, 5, 10, 11, 12, 13, 14, 15 and 18 — are vulnerable to the lack of access to water. The factors responsible for the high vulnerability

includes unsustainable water sources such as water dispensing tankers, water source situated outside the premises, irregular supply even when water is supplied just once a day.

Access to safe and affordable water is critical during hot weather. Since women and children fetch water most of the time, better access will reduce their exposure to extreme heat.

3.3.5 Electricity

It is observed that Rajkot has a very good electricity supply. All nine hotspots fall under the category of low vulnerability. All wards have uninterrupted electricity supply with very few outages even during summer.

3.3.6 Health

All nine wards are highly vulnerable in terms of access to health services. This is due to a lack of an efficient government run health care facility in the vicinity. Longer distances from the nearest government health centre and substantially low coverage of health insurance results in high susceptibility during high heat alert days. Most of the households are out of the health insurance network because they are not able to afford it, and have low awareness of government sponsored schemes such as Ayushman Bharat Pradhan Mantri Jan Arogya Yojana.

3.3.7 Awareness

All the wards surveyed demonstrated poor awareness of heat stress even though they were all highly vulnerable to heat stress. Households are not familiar with the term heat stress and its signs and symptoms. There is practically no awareness of the medical facilities offering treatment for heat stress nor are they aware of the adaptive mitigation strategies adopted by the ward-level administrative bodies of the government, the urban local bodies (ULBs).

3.3.8 Heat stress symptoms

Except for ward number 10 all the remaining eight wards show high vulnerability to heat stress symptoms. The households are highly susceptible to heat exhaustion. Whereas most common heat stress effect reported among men were heat rash (12%), heat exhaustion (11%) and heat cramps (7%), among the women it was heat exhaustion (19%), heat rash (18%) and heat cramps (8%), indicating higher vulnerability and acuteness of the symptoms among women.

Heat cramps

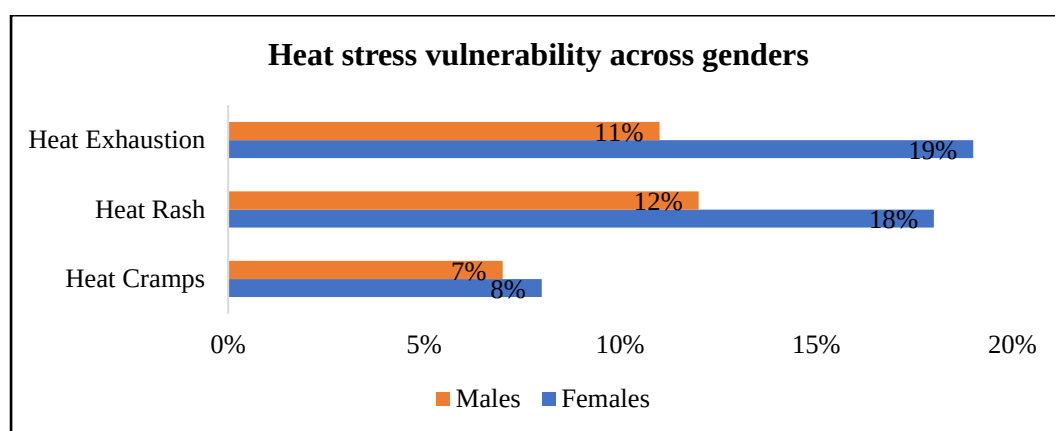
Signs and symptoms of heat cramps include fatigue and painful contraction of muscle groups. Fluid and electrolyte loss often contribute to heat cramps. Prompt treatment usually prevents heat cramps from progressing to heat exhaustion.

Heat exhaustion

Common symptoms of heat exhaustion include heavy sweating and a rapid pulse caused by overheating of the body. It is one of three heat-related syndromes – heat cramps are the mildest and heat strokes the most severe.

The survey figures are evidence that heat stress is a growing public health concern.

Figure: Heat stress vulnerability among women and men



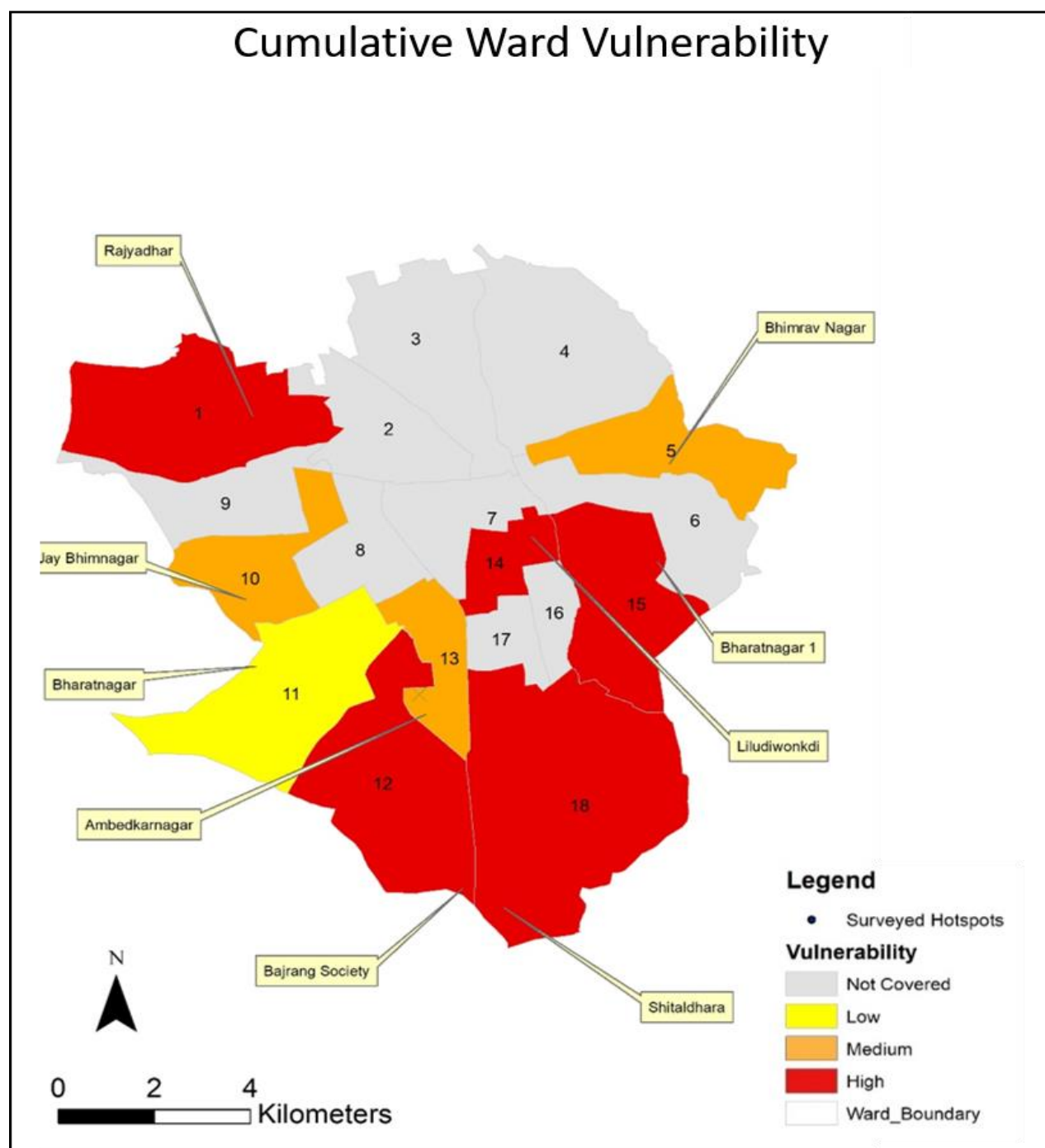
3.3.9 Cumulative vulnerability

The cumulative ward-wise heat stress vulnerability analysis indicated that nearly five wards in Rajkot are highly vulnerable with very little basic amenities available to the community to cope with heat stress.

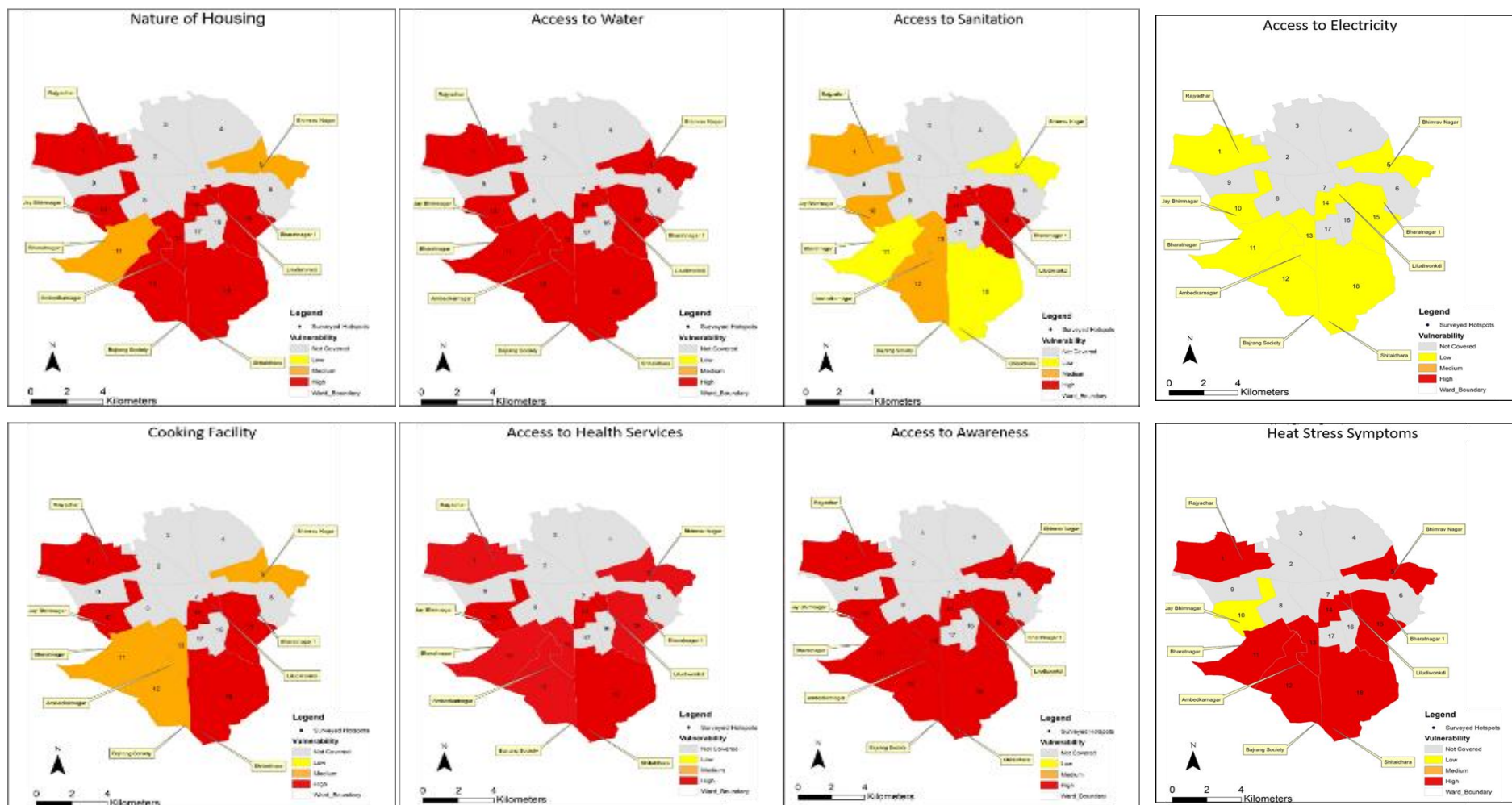
3.3.10 Ward level vulnerability at a glance

The maps below depict how vulnerability is distributed across the nine wards in the heat hotspots. The set of eight maps relate to the index of factors based on which the household surveys were conducted.

Map: Table 4.1: Cumulative vulnerability at the ward level



Vulnerable wards	Wards Number	Total
Low	(11)	1
Medium	(10 , 13 , 5)	3
High	(14 , 15 , 18 , 12 , 1)	5



Map: Factors impacting vulnerability in various wards of Rajkot city

4 Heat Impact Assessment

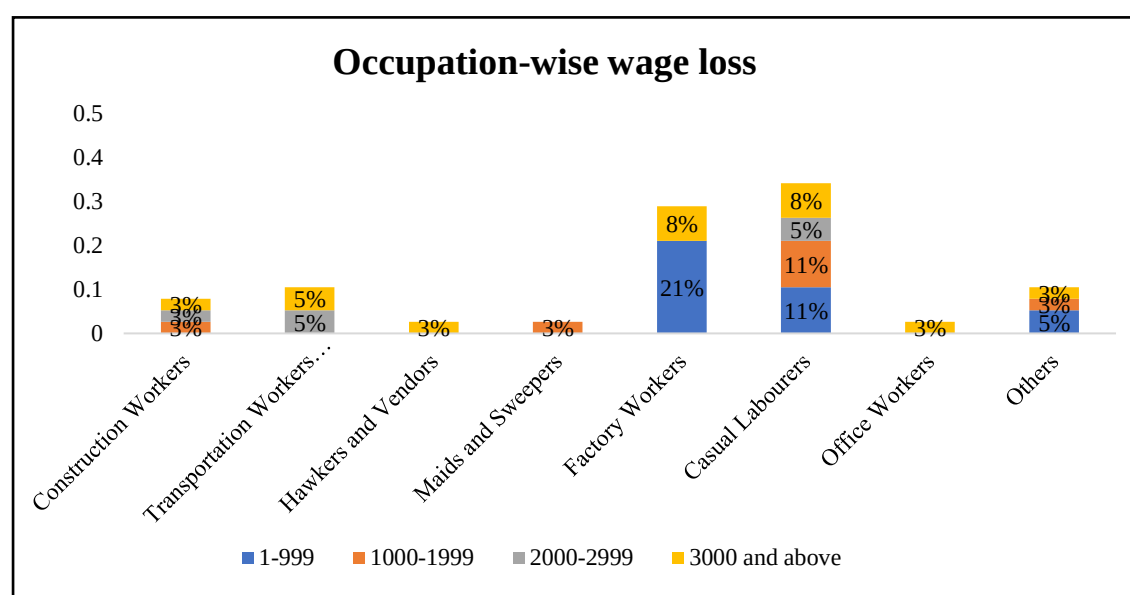
4.1 Impact of extreme heat on income

People vulnerable to extreme temperatures are those who have to remain outdoors for work all day long and have limited options to protect themselves, for example, street vendors, beggars, shopkeepers, traffic police, auto-rickshaw drivers. People who stay at home are equally vulnerable to heat stress because of the material used for roofing and walls absorb heat thereby intensifying the room temperature. In several wards there was very little ventilation in the houses. To make matters worse, many households with no windows use wood as cooking fuel inside the house.

Under these circumstances, both outdoor workers and people who work at home, mainly women who are homemakers, suffer loss of productivity during extreme heat events. Poor coping mechanism and a lack of measures to combat the effects of heat wave frequently results in health problems such as diarrhoea, heat stroke, heat rashes, dehydration, dizziness, among others. And this in turn leads to loss of wages and productivity.

Casual labourers are the ones most affected by high heat days. Workers on daily wages (35%) topped the list of people who reported wage loss owing to high temperatures, followed by factory workers, who due to prolonged working hours often experience heat exhaustion. Hawkers, house maids and office workers were among the least affected because they are in a position to avoid exposure to the sun. The average wage loss in the city falls under the category INR 1-999 followed by INR 3,000 and above. The occupation with average wage loss in the category 1-999 mostly comprises daily wagers while those in the category of above 3,000 have their own small businesses.

Figure: Loss of income according to occupation



4.2 Gender-wise wage loss

It was observed that mostly men (75% out of the total) suffered a wage loss due to heat. Most women lost wages in the INR 1-999 category while men's losses were in the INR 3,000 and above category. The average monthly wage loss among women was INR 200 while among men it was INR 2,000. This significant difference between men and women in terms of wages lost are due to two main reasons: one, a large number of men are in the business sector and have higher incomes and also higher losses during high temperatures; two, a large number of women work as housemaids – their exposure to heat is limited since they spend most of their working day in comfortable homes with better coping measures.

4.3 Productivity loss

With the loss in wages the productivity of the individuals is also highly affected. While a majority (64%) of the working population reported a loss of 1 to 5 working days in a month, there were those (21%) who incurred a heavy loss of 10 to 15 days in a month. Male workers suffered a greater loss (64%) as compared to women. The maximum productivity loss in terms of number of days they stayed off work was reported among factory workers (26%) followed by casual labourers (25%).

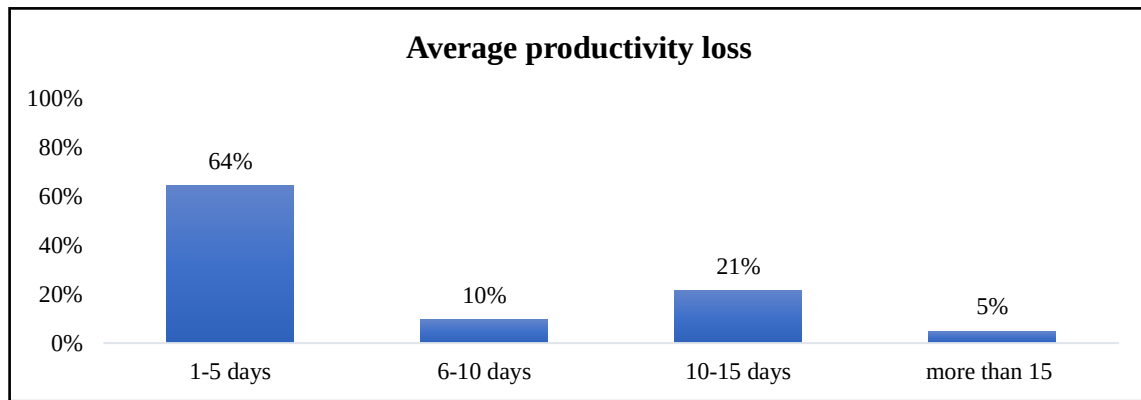


Figure 4.4: Average productivity loss

4.3.1 Loss of productivity and its impact

Loss of productivity can have a spiralling effect. Rising absenteeism at the workplace owing to the health impact of heat stress on a large scale can impact the country's GDP even. It also impacts the family's economy. Other than the medical expenses there is wage loss of the person ill and the caregiver. This means if the husband is absent from work for health reasons, there is loss of productivity for the wife too.

Loss of productivity in the case of a woman implies that household chores are affected too. Children may have to skip school because water has to be fetched. Since the timing is uncertain, families keep checking the water source periodically in order to catch the water when it comes. Of course, when the water does start gushing out finally, there is a buzz in the neighbourhood.

4.4 Summary of survey findings

The broad findings of the survey show that the urban poor are vulnerable to extreme heat for several reasons. A large number of dwelling units in the sample had poor ventilation with cramped interiors, and in several cases with a roof made of heat trapping material like galvanized iron or asbestos.

In Rajkot several poor families live in high-rise buildings. The surveys found that people living on higher floors in small flats with poor ventilation and heat trapping construction material were more vulnerable to heat related impacts. People with disabilities and chronic diseases are however the worst sufferers.

As for drinking water supply, many families did not have piped water at home. They fetched water from a public tap or water supplying tankers, which exposed them to extreme heat during summer.

A critical factor that emerged from the surveys is that households have very little knowledge and awareness of the adaptive and mitigation strategies adopted by the local administration. In a scenario where there is low coverage of health insurance, the government health system needs to recognize heat stress as a public health issue requiring both outreach and facility level services. There is also a great need to build awareness and disseminate information related to extreme heat and how to cope with it. While the impact of heat stress is visible through the sectors, cities can combine sectoral initiatives with a well-defined and coherent framework that ties together both mitigation and adaptation. These provide entry points to combat heat stress in specific sectors of health, housing, infrastructure, and services.

The poor people living in the wards surveyed are more vulnerable to extreme heat events due to their poor coping mechanisms. The night-time outdoor microclimatic conditions along with poor housing structure and poor access to services make it extremely difficult for people to cope with heat stress. The city's heat hotspots undergo significant rise in temperatures as compared to the rest of the city. These areas are most likely to have a higher number of people whose health is affected during a heat wave particularly children, women, the elderly, and construction workers. Identification of vulnerable groups is necessary for greater efficiency in prioritizing treatment of heat related illnesses.

4.4.1 Impact of extreme heat on women

Men, women, pregnant women, children and elderly have different levels of endurance for heat stress. During the surveys male respondents mostly reported heat rash while majority of the women reported heat exhaustion, heat rash and heat cramps during the summer months. Most children and senior citizens also complained of heat rash as well as heat exhaustion. This indicates greater vulnerability of children, women and senior citizens.

Extreme heat also unleashes increased aggression against women. Prolonged hot spells extract a heavy toll, starting with sleep deprivation. People living in cramped houses with heat-emitting walls and roofs lose sleep for days together managing with barely one or two hours of shut-eye at night. This affects . The rising heat also causes more aggression.

As rising temperatures increase aggression, there are more incidences of interpersonal violence, including domestic violence. Poor women are at the receiving end of rising

aggression on a regular basis when it is time to fetch water from water-dispensing tankers. Already precariously poised, trying to balance several buckets and pitchers, they are made more vulnerable because of verbal abuse, and pushing and jostling.

4.4.2 Extreme heat affects women more

Extreme heat affects women differently. Since heat wave conditions are prolonged, the burden of expectation on women to perform chores, like for example cooking sitting in front of a stove fire continuously for many hours, or fetching water from a public water source around noon, increases vulnerability to extreme heat.

More women in the sample households stayed indoors for longer hours compared to men, but this did not make them less vulnerable. Vulnerability is higher within the house because factors like poor ventilation, congestion, and building material used, all trap heat and can cause heat-exhaustion and heat stroke.

It needs to be recognized widely that a pregnant woman is more likely to suffer heat exhaustion or heat stroke because her physiological system has to work harder to cool down her body and the foetus. Pregnancy also makes a woman more prone towards dehydration and so is the case with lactating women. Whether pregnant or not, extreme heat puts women and girls at constant risk of urinary tract infection. Not surprisingly therefore there is an increasing number of heat-related illnesses and stillbirths.

Extreme heat affects everyone, but women, small children and the elderly, particularly older women are more vulnerable. At present, policy does not even link morbidity and mortality with extreme heat. Extreme heat and lack for proper hydration by lactating mother can dry up breastmilk, leaving the newborn severely malnourished and dehydrated. The fact is extreme heat affects women differently and at many levels.

4.4.3 Impact on nutrition

Other than general loss of appetite during periods of extreme heat, diarrhoea and vomiting also take a toll on the nutrition status of families particularly children.

Family nutrition is also compromised in households that do not have a refrigerator because preserving food becomes difficult – the intake of fresh vegetables is reduced considerably as a result.

4.4.4 Dehydration

Dehydration and summers go hand in hand. Excessive heat or heavy sweating causes loss of body fluids and electrolytes and can be replenished with simple measures like regular intake of lemon water with salt, an occasional banana for potassium. But dehydration frequently leads to complications and needs medical intervention because of a lack of awareness.

4.4.5 Awareness building

Since women in general are custodians of the family's health and nutrition there is a need to reach out to them to make them aware of simple coping mechanisms. Building such awareness will prevent simple cases that can be managed at home from progressing to medical complications requiring a doctor's intervention. A little awareness can therefore save not only expenses, but also loss of productivity.

4.4.6 Interpersonal communication

Studies and interventions in maternal health have shown that person-to-person communication on a regular basis is most effective. Several platforms exist that can disseminate messages related to heat once there is a policy push. The health department already has in place urban ASHAs (accredited social health activists)¹⁴, who are community health service providers at the ward level. This health worker works in close coordination with the anganwadi worker (AWW) who runs the nutrition programme for small children and pregnant and lactating women from a childcare centre called the anganwadi centre (AWC). A municipality ward may have more than one AWC depending on the population. In addition, every ward has a Mahila Arogya Samiti (MAS), a women's health group.

4.4.7 Sanitation

A large number of households in the survey had no access to a toilet. The vulnerability of women, children and the elderly, who are already the more vulnerable, increases manifold when they have to walk a distance for defecation in the open.

4.4.8 Water supply

¹⁴ Ashas are women community health workers put in place by the National Health Mission.

Only a small percentage of households in the survey have the supply network pipes reaching their dwelling unit. That is the reason several households fetch water from sources like public taps, water-dispensing tankers that bring water to the neighbourhood, nearby streams or dug wells.

Women in some of the sample areas have to run the house with water supply that comes once in many days. Their vulnerability increases sharply because they have to fetch water from many sources, walking in the hot sun with containers. Sometimes children have to miss school because they have to fetch water.

4.5 Other areas where policy intervention is needed

Awareness building is the area where a major policy thrust is acutely needed. That extreme heat is the cause of several health issues needs to be understood on a wider scale.

The survey brought to light that there are several areas where awareness building is needed. Some of these are:

4.5.1 Health insurance

An overwhelming majority of the households surveyed do not have health insurance. Most of the households are out of the health insurance network because they are not able to afford it, and have little awareness of government sponsored medical insurance schemes such as Ayushman Bharat Pradhan Mantri Jan Arogya Yojana. To improve access to health insurance, the government needs to promote and publicize its insurance schemes meant for the below-poverty-line population.

ASHAS can be roped in to build this awareness.

4.5.2 Treatment for heat stress

Awareness of heat stress is low as also the measures. An overwhelming percentage of the sample households are not aware of the medical facilities offering treatment for heat stress. Stakeholders need to come together to ensure that information is disseminated among households about the impact of extreme heat and ways to mitigate it.

5 Heat Threshold and Early Warning System for Rajkot

Since the climate varies from location to location, the concept of an extreme event (weather or climate) and its threshold often vary from location to location. In other words, what is considered to be the extreme value of a given climate feature at one location can be considered to be within the normal range at another location. Certain meteorological thresholds (i.e. two or more consecutive days at a maximum temperature above a certain value) are used to measure heatwave extremes and cause alert systems. Some action aims to develop to protect vulnerable population have also been drawn up on the basis of such thresholds and by including details on threshold for triggering the early warning system could be initiate. The city-specific threshold determination is mandatory as it is not available for many cities. There is variability in what threshold would trigger an action plan and how much delivery time taken by the organizations and the “at risk populations” are provided to prepare for it. In a number of heatwave early warning systems (HEWS), the forecasted threshold must exceed on a number of trigger a warming.

Methods: The warming impact particularly sever in tropical areas, which mainly consist of developing countries including India (Sathaye et al., 2006). This is the reason behind the selection of western part of India. Average summer mean maximum temperatures of Maharashtra have increased by +0.02* between 1951-2010 and shown significance at 95% level (Rathore et al., 2013). To develop thresholds for the selected cities, weather data – daily maximum and minimum temperature and humidity were obtained from Indian Metrological Department (IMD) for summer months between March to June (2003-2017) (Golechha et al., 2021). In their jointly published guidelines for developing a heat health warning system, WMO and WHO advised using percentile-based thresholds at the 90th and 95th percentile values to activate early warning systems in cases where mortality or morbidity data are unavailable and only meteorological data are (McGregor et al., 2015). For this action plan, we determined the 75th, 85th, and 95th percentile values for the following thresholds: (1) a common seasonal threshold for the entire summer (March to June); and (2) separate thresholds for each summer month, i.e., the 75th, 85th, and 95th values for March, April, May, and June separately for Rajkot city where the percentile-based method was used.

Threshold analysis for Rajkot city

Heat alerts based on thresholds determined by IMD, as an additional means of communication by using the following color signal. Red alert declared when temperature raised above 45° C for Rajkot city.

Heat-Health Temperature Warning for Rajkot		
Yellow alert	Hot day advisory	41.1°C- 43° C
Orange alert	Heat alert day	43.1°C – 44.9° C
Red alert	Extreme heat alert day	Above 45° C

Heat Alert Thresholds for Rajkot City

Below table shows, 75th, 85th and 95th percentile values of maximum temperature to determine thresholds for April, May and June and common seasonal threshold for the summer months mentioned below table for Rajkot city.

		Yellow alert (75% percentile)	Orange alert (Heat alert day) (85% percentile)	Red alert (Extreme heat alert day) (95% percentile)
Rajkot	April	42.2° C	42.7° C	43.5° C
	May	42.9° C	43.5° C	44.2° C
	June	42.6° C	43.1° C	44.2° C
	Common seasonal threshold for entire summer (March to June) months			
		75% percentile	85% percentile	95% percentile
		42.7° C	43.2° C	44.0° C

* Source: IMD Meteorological data 2003 to 2017

Month-wise and seasonal Thresholds for Heat wave warning for Rajkot

Sathaye, J., et al. 2006. Climate change, sustainable development and India: Global and national concerns. Curr. Sci. 90, 314-325.

Rathore, L.S., et al. 2013. State level climate change trends in India. India Meteorological Department (IMD), 2013. Meteorological Monograph No. ESSO/IMD/EMRC/02/2013.

Golechha M., et al. 2021. Threshold determination and temperature trends analysis of Indian cities for effective implementation of an early warning system. Urban Climate. 39, 2021.

McGregor, G.R., et al. 2015. Heatwaves and health: Guidelines on warning system development. WMOP.

https://www.who.int/globalchange/publications/WMO_WHO_Heat_Health_Guidance_2015.pdf (accessed 14 May 2021).

6 Climate adaptive heat action plan

6.1 Strategies of the plan

The climate adaptive heat action plan is designed to improve the management of heat-related risks in Rajkot. At its basis is spatial distribution of the city's heat hotspots and the poor people living in these heat islands. The reason is simple: these heat islands are warmer compared to the other parts of the city and the urban poor have very few coping measures to combat extreme heat. The plan is also gender-sensitive, because women have to cope with much more during high temperatures for reasons of body physiology and cultural construct – domestic chores like cooking continuously before a stove fire, fetching water from a source outside the house, among others, increase their exposure to high heat. The overall purpose of this plan is to support the city municipal corporation to prioritize and adopt strategies that will help Rajkot become a climate resilient smart city.

This heat action plan is also comprehensive. It provides a framework for implementation, coordination between municipality departments and evaluation of extreme heat response and guides mitigation and adaptive measures to prevent loss of life and productivity. The plan's primary objective is to alert populations at risk of heat-related illnesses where extreme heat conditions either exist or are imminent so that people take appropriate precautions. The plan therefore helps to bring together all stakeholders for a citywide strategy in enforcing preventive, mitigation and adaptive measures to check heat-related debility.

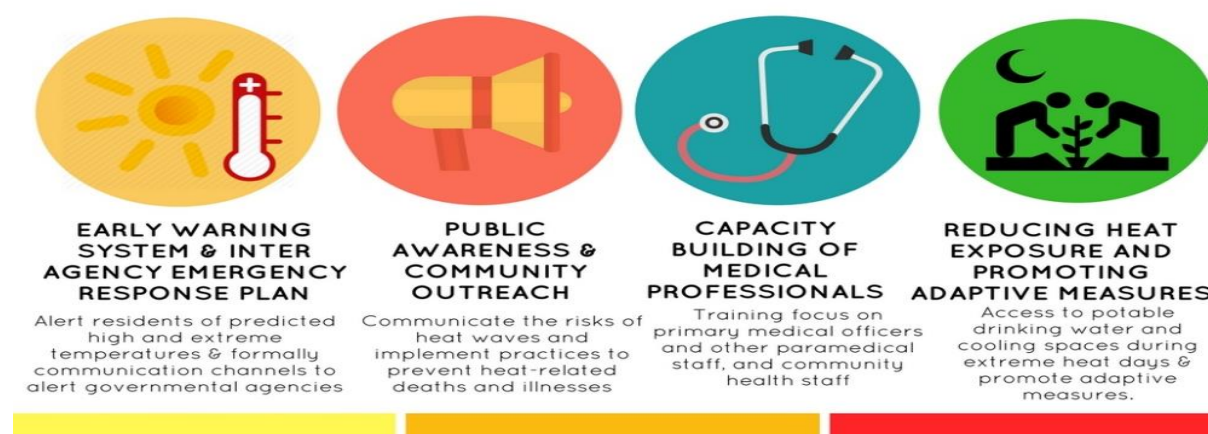


Figure: Key steps to prepare for heat waves (Source: Ahmedabad Heat Action Plan, IIPH-G)

6.1.1 Preparedness

The following steps are key to preparing for heat waves and prevent loss of health and productivity.

1. *Building public awareness and community outreach* to communicate the risks of heat

waves and adopt measures to prevent heat-related deaths and illnesses. Public messages through various media such as fliers, radio and television spots will serve as a timely reminder for the community to stay prepared and take appropriate measures when extreme heat strikes. This reminder also needs to go out via text messages and social media applications such as WhatsApp or Instagram. Special efforts have to be made to reach out to vulnerable populations, including women, through interpersonal communication as well as other means of outreach.

2. *Setting up an early warning system and inter-agency coordination* to alert residents of forecasted high/extreme temperatures. The IMD already shares a daily five-day forecast with the Heat Action Plan nodal officer of the Rajkot Municipal Corporation during summer. What is needed is to create formal communication channels to alert government agencies, the meteorological centres, health officials and hospitals, emergency responders, like RMC's health department officials, city disaster management cells, local community groups, and media.
3. *Capacity building of health professionals* is needed so that they are able recognize and respond to heat-related illnesses, particularly during extreme heat events. Many are already doing this, but a formal training session will help to sharpen diagnosis as it will remove any uncertainty about health issues extreme heat can cause. Who will need such capacity building on a priority basis – primary healthcare medical officers, paramedical staff, and community health workers (ASHA). It will help improve prevention and management of heat-related cases.
4. *Reducing heat exposure and promoting adaptive measures* like mapping of high-risk areas of the city, increasing outreach and communication of preventive measures, and setting up intermittent drinking water points and cooling sheds for the benefit of pedestrians and commuters. Collaboration with NGOs is a means to extend outreach and communication among the city's most at-risk communities.
5. *Other measures can include*
 - A heat emergency response plan
 - Collaboration with NGOs and civil society to develop 'cool public places' particularly bus stands, building temporary shelters, and setting up drinking water points

6.1.2 Adaptation and mitigation measures

The rising number of heat waves is marked by a large number of heat-related illnesses. Many of these turn serious, causing death even. The state of disaster that heat waves unleash plays itself out in hospitals mostly, as heat stress is causing an increasing number of medical emergencies. This leaves the healthcare delivery system struggling to cope with the heavy caseload.

Such circumstances call for deployment of a quick response plan that deals with medical emergencies without overwhelming the system. This adaptation-and-mitigation strategy therefore lays down measures to get hospitals and other health service providers to prepare for the stream of patients suffering from heat-related illnesses.

It would require planning, coordination, capacity building, monitoring, appraising, and acting in accordance with the procedures laid down along with collaborative efforts of all stakeholders. A hospital emergency preparedness plan should generally take into account all aspects of heat waves including protocols to be followed before, during and after a heat wave.

6.1.3 Steps to take before the season sets in

1. Create and implement gender-sensitive heat health guidelines on how to recognize and treat heat stress, heat exhaustion, and heat stroke.
2. Set up a steering committee to supervise operations, monitor emergency preparedness, manage the heavy caseload of people affected by heat waves, and post-heat wave evaluation.
3. Ensure adequate stock of IVs, ORS, and other medicines that may be needed in bulk.
4. Ensure availability of beds, especially in emergency departments and special wards for heat related illnesses, especially among women.
5. Set up cooling mechanisms in wards. Since the temperature build-up is the highest on the top floor relocate maternity or neonatal wards if they happen to be here. Move patients with heat-related health issues to cooler floors/parts of the building.
6. Measure wards' morbidity and mortality rates before and after location change to evaluate the effectiveness of the intervention.
7. Increase staffing – medical doctor and paramedical staff to ensure preparedness in case of increase in admissions.

8. Extensive use of IEC material, including posters and fliers, and provide advisories to prevent heat stress.
9. Develop training modules or trainings for healthcare providers, ward leaders, and paramedics on extreme heat and health, as well as specific heat case management and diagnosis, especially during heat waves.
10. Organize training of trainers for primary healthcare medical officers so they can offer heat-specific advice (symptoms, diagnosis, and treatment including self-monitoring hydration) to their medical staff.
11. Conduct workshops for frontline health workers (Asha, Anganwadi workers, community health workers) to increase gender sensitive outreach and community-based surveillance for heat illnesses among poor communities. Frontline health workers need to be provided with information tools that is pictorial as these will help them to counsel the community especially women.
12. Collaborate with the medical service providers/research institutes to train emergency service professionals on responding to extreme heat emergency cases.
13. Since newborns are particularly vulnerable to heat stress, increase heat stress outreach and education for women in maternity wards before they leave the hospital. Coordinate with National Health Mission to revise Asha training module on post-natal newborn care to include management of heat wave conditions.
14. Update heat wave monitoring and management protocols and programmes, including tracking of daily heat-related data that is gender disaggregated.

6.1.4 Steps to take during the season

1. Adopt gender specific heat-focused examination procedures at local hospitals and Urban Health Centres. Train Ashas on these protocols. Make it a routine to examine admitted patients for signs and symptoms of heat-related illnesses adding a brief procedure during the peak summer months. Basic information of such patients should be noted to maintain a record of their location, occupation, sex, and socio-economic status to build a database of city's residents most vulnerable to extreme heat.
2. Adapt pharmacological treatments according to Standard Treatment Guidelines (STGs). Gender aspects should be given due consideration.
3. If possible, postpone non-emergency hospitalizations and surgeries.

4. Ensure high risk patients are placed in rooms with air-conditioning; less critical patients should at least have access to an area with air conditioning/air circulation during the hottest hours of the day.
5. Increase liquid oral and intravenous intake of patients.
6. Modify diet with increased intake of fruits and vegetables.
7. Adjust patient bed and personal clothing according to need.
8. Start special and adequate health and social assistance for hospital discharge of high-risk patients, especially new mothers with babies or postpone discharge till heat wave subsides.
9. Ensure availability of adequate number of vans for medical mobile units in high-risk areas.
10. Maintain records of heat-wave patients according to the government prescribed monitoring sheet and report to the municipal corporation on a daily basis.
11. Expedite recording of cause of death mentioned/documented in death certificates.

6.1.5 Steps to take after the season

1. Share final gender-disaggregated data on hospital admissions as per indicators set for reporting during heat wave with the municipal corporation.
2. Give feedback in annual evaluation of heat action plan.
3. Prepare a set of key learnings during heat waves to build on institutional memory and share it with other stakeholders.

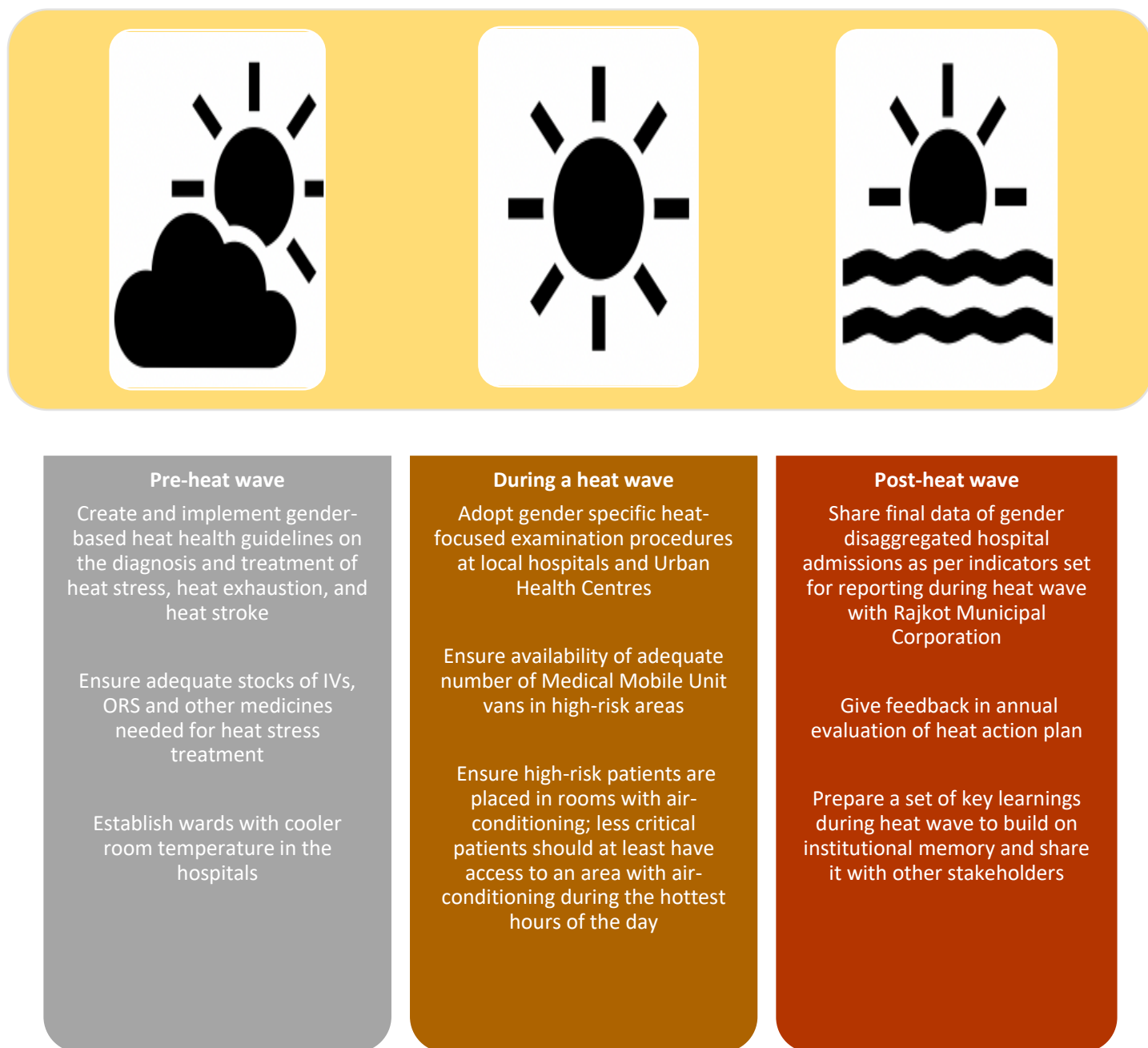


Figure: Broad steps to take before, during and after peak summer

6.2 Heat Action Plan

Rajkot Municipal Corporation has been implementing the heat action plan since 2017 and has taken extensive measures for implementation of the same under the leadership of Municipal Commissioner, RMC. Three-pronged with short-, medium- and long-term measures. It is

important to understand what these measures are in order to build on them and put in place a climate adaptive heat action plan.

6.2.1 Short- and medium-term measures

Awareness campaigns

- Hoardings, posters, to be displayed by LED TVs at various locations, distribution of fliers
- Awareness workshops for people whose occupations entail heat exposure – traffic police, hawkers, street vendors, construction workers and school children

Mitigation measures

- Keeping gardens, cooling shelters and other possible cooling centres open with provision of drinking water
- Availability of drinking water and sheds at open construction sites
- Pilot project on painting roofs white to lower the room temperature of the house and distribution of gunny bags in slums and squatter settlements. These are for lowering temperatures in shacks by drenching them with water and placing on tin/asbestos roofs.
- Setting up water and ORS points at various places – construction sites, bus stands and other public places, on the route of processions and at rallies during summer
- Distribution of cooling vests to on-duty traffic police personnel. Cooling vests help regulate body temperature during hot days outdoors.
- Water tankers to be made available on call in slums during orange/red alert days
- Developing and provision of night shelters for migrants
- Provision of 24x7 electricity supply during the heat waves

Early warning communication

- SMS and WhatsApp messages for early warning to citizens, NGOs, citizen welfare groups, construction contractors
- Roving public announcement on mikes across the city on orange and red alert days – a day before and once more early on the forecasted day
- Press releases and campaigns on radio, TV and websites

Medical preparedness

- Stocking ORS and cool packs at the health centres and readiness with cooling and rehydration as well as shock management treatments
- Medical camps on day of red alerts at heat hotspots

Monitoring and warning mechanism

- Improve the existing the monitoring and warning mechanism
- Procure and install automated weather stations in each ward
- Maintain ward-wise record of heatstroke cases, daily mortality with cause of death as well as daily hospital admissions of all-cause cases and heat-related causes
- Monitor and analyse morning temperatures recorded at weather stations and issue early warnings
- Acquire ward level temperature forecasts and heat wave warnings from IMD

Ward level heat stress management

- A ward level heat stress management committee needs to be formed comprising:
 - Ward councilors
 - Health officials
 - Ward officials of the municipal corporation
 - NGOs and the research institutions working at ward level
- Develop ward level heat action plans

6.2.2 Long-term measures

- Heat alerts and emergency response plan needs to target vulnerable groups, high-risk areas and incorporation of the same in the City Development Plan. Planned development is required of urban areas ensuring appropriate amenities are available to all residents in every location.
- Heat insulation and building standards need to be strengthened by improving building bye-laws along with increased heat tolerance for new infrastructure. Building bye-laws can have components of passive ventilation and cool-roof technologies and made mandatory in more vulnerable areas.
- Identify locations and build shelters and sheds for the urban poor.
- There is a need to collate and document indigenous knowledge to promote traditional coping mechanisms.
- Carry out capacity building at the community level through awareness campaigns and outreach programmes and by communicating risks associated with heat stress – its impact on health, livelihood and productivity and ways to mitigate the same.
- Initiate research on micro-climate and corroborating the need to monitor temperatures in urban areas. Policy level intervention to retrieve natural eco-systems and natural shelters.
- Encourage vertical greenery, roof gardens as means to improve the urban landscape with alternative methods to bring down the temperature. Greening infrastructure can be an effective method to cope with heat stress. Urban forests have been found to be effective. A combination of shading, reduced heat build-up through judicious use of building material, humidity and wind management can provide heat refuge at street levels.
- Encourage investing in water bodies, fountains in areas of mass presence and promote greening of city spaces along with improving green transport and energy systems.

6.2.3 Capacity building

A medical stakeholders' training-cum-orientation workshop was organized for healthcare professionals. The training aimed at orienting healthcare professionals of Rajkot on the heat action plan and enhancing their capacities for proper and inclusive management of heat-related illnesses. More than 50 doctors and public health professionals underwent the training; it not only covered heat stress and protocols for heat-related diagnosis and treatment, but also overall preparedness for prevention and management of heat stress.

6.3 Heat wave advisories

Heat wave conditions can result in fatal physiological strain. To minimize the impact of heat waves on health it is best to practise caution. The following are some dos and don'ts that serve well.

Dos and don'ts for heat waves

- ✓ Follow weather forecasts and advisories on radio, TV, newspapers.
 - ✓ Drink water often, even if not thirsty.
 - ✓ Wear lightweight, light-coloured, loose cotton clothes; use protective goggles, umbrella/hat, shoes or chappals while going out in the sun.
 - ✓ While travelling, carry water with you.
 - ✓ If you work outdoors, use a hat or an umbrella and also use a damp cloth on your head, neck, face and limbs.
 - ✓ Take ORS along with homemade cooling drinks like lassi, torani (fermented rice water), lemon water, buttermilk, to rehydrate the body and replenish mineral salts.
 - ✓ Learn to recognize the signs of a heat stroke, heat rash or heat cramps such as weakness, dizziness, headache, nausea, sweating and seizures. If you feel faint or ill, see a doctor immediately.
 - ✓ Keep animals in the shade and give them plenty of water to drink.
 - ✓ Keep your home cool, use curtains, shutters or sunshade and leave windows open at night.
 - ✓ Use fans, damp cloth and cold water bucket-baths frequently.
 - ✓ Provide cool drinking water at workplace.
 - ✓ Caution workers to avoid direct sunlight.
 - ✓ Schedule strenuous jobs for cooler times of the day.
 - ✓ Increase the frequency and length of rest breaks for outdoor activities.
 - ✓ Pregnant women and workers with a medical condition should be given additional attention.
- ✗ Do not leave children or pets in parked vehicles.
 - ✗ Avoid going out in the sun, especially between noon and 3.00 pm.
 - ✗ Avoid wearing dark, heavy or tight clothing.
 - ✗ When the outside temperature is high, avoid strenuous activities especially between noon and 3 pm.
 - ✗ Avoid cooking during peak heat hours. Open doors and windows to ventilate cooking area.
 - ✗ Don't consume alcohol, tea, coffee and carbonated soft drinks as these drinks dehydrate the body.
 - ✗ Avoid high-protein food and do not eat stale food.

6.3.1 Special Strategies for Vulnerable groups

Communication strategies • Temperature and forecast display • Proper communication during heat alerts • IEC activities that reach all the vulnerable groups • Helpline centres	Cooling strategies • Public access to potable water and cool spaces • Development of public parks • Shaded bus stops • Community resource map • Greening and tree plantation campaigns • Cool roof campaigns	Occupational strategies • Outdoor labourer education • Labour law enforcement and coordination • Cooling incentives for local/outdoor businesses
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7 Implementation of the action plan

7.1 Creating a framework for coordinated approach

As extreme heat affects multiple aspects of life its coping measures too need a multi-sectoral approach. For smooth implementation of a climate adaptive heat action plan on an annual basis RMC needs to institutionalise a heat action component in the yearly plan of several government departments. Through a series of workshops RMC needs to bring about a well-coordinated action plan on the ground. The purpose of these workshops is to help the following departments to formulate an annual heat action plan of their own and make it official by including it in the coming year's budget of the department. The key departments RMC needs to coordinate with are:

- Drinking water and sanitation
- Education Department
- Electricity
- Transport
- Labour
- Health and the National Urban Health Mission
- Women and child development department that runs the child development (anganwadi) centres in the community

Once RMC brings the above government departments to formulate and own an annual heat action plan it will bring these different sectors on the same page with regard to mitigation measures for extreme heat. Such a coordinated approach will pave the way for smooth implementation of a multi-sectoral heat action plan at the ward level.

Since extreme temperature triggers heat related health issues on a wide scale a lot depends on how efficiently RMC brings about convergence at the ward level with the health department, the urban programme of the National Health Mission (NHM) and the department of woman and child development that is responsible for the child development centres in the municipal wards. The frontline workers of these three government agencies already work in coordination to achieve goals related to maternal and child health. RMC needs to add on heat stress to their ongoing awareness building and health programmes.

While the ANM, a trained paramedical, is based in the primary health centre, the Asha and AWW are both based in the community. The AWW is based in the aanganwadi centre, the child development centre located in the ward. The ANM, ASHA and AWW regularly conduct awareness sessions related to maternal and child health, and nutrition. RMC now needs to take the necessary steps to include impact of extreme heat on the health of mother and child. To be able to do so the municipal corporation needs to work with NHM to bring in 'impact of extreme heat' as a chapter in the pictorial flip books supplied to ASHAs to help them conduct awareness sessions. Since there are flip books on various topics that cover the health of pregnant women and new-borns, the extreme heat factor needs to be included in all these visual aids.

7.2 Identifying high risk households

Since ASHA and AWW both are based and work in the community, they are best suited to identify households most vulnerable to heat stress. For example, if during her scheduled home visit of pregnant women she notices a household does not have an electric fan, she can bring it to the notice of the ward heat stress management committee. Similarly, ASHA can make a list of the households that are more vulnerable due to tin/asbestos roofs in her catchment area and share with the heat stress management committee. This will enable the committee to arrange for supply of gunny bags to these households.

The vulnerability survey IRADe conducted showed a large number of households are at high risk of heat stress because of the use of firewood as fuel for cooking. Asha can identify these households and share it with the ward heat stress management committee.

RMC however needs to coordinate with the respective departments so that these two community based frontline workers are given proper orientation in identifying signs and symptoms of heat stress and heat vulnerability.

7.3 RMC and primary health centre

RMC needs to work closely with primary health centres to strengthen their outreach programme.

7.4 Learning from other campaigns

At the state level though the government can take a cue from ongoing campaigns and adopt some of the strategies that work. For example, the campaign on nutrition to combat malnutrition in India introduced simple messages in school textbooks and raised awareness at an early age. Civil society led the school WASH (water, sanitation and hygiene) campaign by working among students of classes 6 to 8 to raise awareness on personal hygiene; raising awareness among students also helped reach the message to their families. The government can consider similar policies in relation to coping mechanisms for heat waves.

RMC also needs to work alongside the health department when it prepares its annual budget in order to help formulate the coming year's budgetary allocation for mitigation measures for heat stress. Including the heat stress related action points in the budget is important because it will then make heat stress mitigation part of health policy, and therefore more binding on primary health centres that function at the ward level.

7.5 Coordination with transport department

This is important because a large percentage of the heat vulnerable population are workers who have migrated from other districts of Gujarat. They live in temporary settlements and periodically travel long distance by bus to their village. During peak summer a long journey in a non-airconditioned bus can be harrowing and lead to heat stress. RMC needs to coordinate with the transport department and drinking water-and-sanitation department to make sure the major bus stations and bus terminals have safe drinking water taps and toilets.

RMC also needs to coordinate with the transport department to include setting up of ORS-and-drinking water kiosks at city bus stops in its annual plan.

7.6 Specific advice for occupational groups

The labour department needs to be made aware of the need for advisories tailored for different occupations. The common occupations in the heat vulnerable wards are construction workers, factory workers, street vendors and hawkers, and those that run small businesses on the roadside like tailoring and food cart.

7.7 Toilets, drinking water and electricity for the more vulnerable

RMC will need to work with drinking water and sanitation department to take its critically important services to the more vulnerable among the heat vulnerable population of Rajkot. Every year there are new temporary settlements to be seen around large construction sites or near roads and highways being built. These settlements are a cluster of dwelling units of a basic nature made of plastic sheets and asbestos or galvanised iron sheets serving as a roof.

RMC and the department could work together to factor in temporary leach-pit toilets the kinds that are used in large fairs and religious gatherings like the Kumbh Mela. Leach pit toilets direct the solid and liquid waste to a pit and allows nature to take its course to convert it into compost. The department erected millions of such toilets, both temporary and permanent, during the nationwide sanitation campaign (2014-2019) to end open defecation.

RMC also needs to coordinate with the drinking water and sanitation department to make budgetary allocations for daily supply of drinking water to these temporary settlements who are more vulnerable to heat because of a lack of access to services.

Similarly, RMC needs to coordinate with the electricity department to work out a way to provide a temporary connection for a light source and an electric fan to each shack in the temporary settlement. The electricity department will need to own this intervention by including it in its annual plan.

7.8 Identifying new settlements

Perhaps the first step towards planning for the upcoming summer season RMC needs to do a roof-count of vulnerable households. It needs to conduct a survey every year to identify all temporary settlements – old and new – so that they are not left out of the annual heat action plan. This exercise needs to be carried out even before beginning to plan for the summer ahead.

7.9 Raising awareness

As discussed earlier, RMC needs to work with the health department and the National Health Mission to include heat stress into the communication tools Ashas use to raise awareness on health of pregnant women and children. These two groups are also among the more vulnerable to heat stress. Since these awareness sessions encourage participation of men, they provide a good platform to teach the community about heat stress and its early detection.

RMC needs to work with civil society and community-based organizations to develop IEC material and raise awareness through various methods, including posters, wall paintings, messages from a roving microphone, and interpersonal communication. Several NGOs working on maternal and child health have their own community health workers, who can be roped in for interpersonal communication on heat stress and its early detection, and coping mechanisms.

Once RMC has worked out its coordination with different departments to include heat stress in their plans the annual heat action plan will then become routine. Here is a suggestion of the possible steps to take, which the municipality department heads will need to fine-tune.

7.10 Annual Heat Action Plan

The annual heat action plan below suggests a roadmap of the steps to be taken, and who will carry them out. The action plan needs to divide steps to be taken into three phases – before the onset of summer which is the preparatory phase, during the hot season and post-summer.

Phase 1: Before the onset of summer (January-February)

This period is devoted to developing early warning systems, communication plans for heat alerts to the general public and training/orientation of health service providers, including PHC doctors, paramedics (ANMs) and frontline workers (Asha, AWW). This is also the phase when the database of vulnerable households needs to be updated.

Phase 2: During peak summer (March to June)

This is a period of high alert, continuous monitoring of the situation, coordination with all the departments concerned, the community and media.

Phase 3: Post-summer (July to October)

During this phase the main task is evaluation and improving the plan based on the lessons learned. It is important at the end of the summer to evaluate the heat-health action plan. Global climate change is projected to further increase the frequency, intensity, and duration of heat-waves and attributable deaths. Continuous updating of the plan is therefore necessity.

7.10.1 Roles and responsibilities in Phase 1: Before the onset of summer (January-February)

RMC Nodal Officer

- Collaborate with Rajkot educational institutes and civil society to create gender-sensitive heat health guidelines on the recognition and treatment of heat stress, heat exhaustion, and heat stroke to reduce and prevent mortality and morbidity. These guidelines need to be distributed widely and in at least three languages – Gujarati, Hindi and English.
- Set up a steering committee to supervise and monitor emergency preparedness, dealing with inflow of patients during heat wave and post-heat wave evaluation. This steering committee will work in close coordination with the ward heat stress management committees, and through them monitor the ward's performance.
- Collaborate with the medical service providers/research institutes to train emergency service professionals on responding to extreme heat emergency cases.
- Convene a meeting of key stakeholders, including representatives of Gujarat State Disaster Management Authority, NGOs, community-based groups, media, health department, private hospitals, departments of labour, water and sanitation, transport, power supply and distribution, private institutions, religious groups. The purpose of this meeting is to respond to extreme heat events (See figure titled Communication Plan When the RMC Nodal Officer Activates a Heat Alert).
- Conduct workshops in schools and in the wards. Engage with Ashas and AWWs in conducting these workshops.
- Engage with civil society to launch awareness campaigns on health risks of heat stress through multilingual pamphlets, posters at vantage locations in hospitals, schools, bare walls in the ward and public and private institutions. Use other media like roving microphones, and interpersonal communication in the community to raise awareness.
- Carry out a survey to identify and draw up a list of temporary settlements and other high-risk areas in the city where people are more vulnerable to heat waves for focused preventive measures.

Media and RMC Press Officer

- Execute campaign and awareness outreach on risks of exposure to high temperature,

heat stress prevention, and tips for health protection during extreme heat events with greater focus on high-risk areas.

- Ensure wide visibility of information and heat communication materials to the public.
- Increase the number of LED screens installed to display daily temperature forecasts for public view.

RMC Health Department

- Coordinate with primary health centres of vulnerable wards to conduct outreach of health services to vulnerable communities. This outreach activity needs to be dedicated to coping with extreme heat and scheduled over and above the ANM's routine outreach efforts.
- Organize training/orientation of frontline workers (ANM, Asha, AWW) on early detection of heat stress and on raising awareness of preventive measures in the community.
- Conduct training workshops for capacity building of doctors, nurses and ANMs to help manage the increased caseload in hospitals and health centres following extreme heat events.
- Conduct workshops for hospital administration personnel to orient them towards updating their admissions and emergency case records to track heat-related morbidity; train them in recording heat stroke/heat stress as the cause of death in certificates, if death is triggered by an illness from exposure. This will create a reliable dataset to analyse epidemiology of illnesses associated with heat stress.
- Orient doctors/nurses/ANMs working in city hospitals and government health centres to adopt heat-focused examination procedures during summer.
- Conduct workshops with primary health centres, community health centres, 108 emergency service personnel, private ambulance service providers and private hospitals and nursing homes to prepare for the onrush of cases that will come with the start of heat waves.

RMC Labour and Employment Department

- Develop a pre-recorded message on what to expect during heat waves, coping mechanisms and preventive measures. Let a person move on a slow-moving vehicle like a battery-operated rickshaw and move from place to place. This person can park at

construction sites and other places where employers, middle managers, workers and owners of small businesses can listen to the pre-recorded message. Create a roster for the places to cover on days of the week. The pre-recorded message needs to be both in Gujarati and in Hindi as well for the benefit of migrant workers.

- Identify high-risk outdoor workers and give them focussed attention in outreach and advocacy.

108 Emergency Service

- Create displays on ambulances to build public awareness.
- Identify vulnerable populations in at-risk areas and be in a state of preparedness to provide immediate relief.

7.10.2 Roles and responsibilities in Phase 2: During peak summer (March to June)

RMC Nodal Officer

- Activate citywide heat alert and response mechanism based on the Department of Meteorology's weather forecast, by notifying key stakeholders, RMC Deputy Municipal Commissioners and the Gujarat state agencies in accordance with the Communication Plan (See figure titled Communication Plan: When the RMC Nodal Officer Activates a Heat Alert).
- Monitor the heat alert level based on the weather temperature severity forecast. Increase in severity level necessitates the Municipal Commissioner to convene a special meeting of key agency leaders.
- Expand access to shaded areas for outdoor workers, slum communities, and other vulnerable populations during a heat alert. This can be done by opening up 'cooling centres,' which can be temples, public buildings, malls, RMC-run temporary night shelters.
- Hold daily meetings to monitor progress during a heat alert.
- Ensure continuous surveillance of temperature data and forecasts for appropriate action.
- In the event of water shortage communicate suspension of all non-essential use of water.
- Engage with religious groups and civil society to hand out water pouches at key public

places and bus stands.

- Communicate local utility protocol to prioritize uninterrupted power to critical facilities (such as hospitals and health centres).

Media and RMC Press Officer

- Issue heat alerts through WhatsApp and SMS platforms utilizing the centralized mobile databases of private sector telecom companies.
- Issue heat alerts to the civil society, social service workers and public via centralized email databases.
- Send direct heat alert messages to private medical practitioners, hospitals and health centres.
- Utilize social media, local radio FM broadcasts to disseminate heat protection tips and high temperature warnings to the city's at-risk populations.

RMC Health Department

- Deploy vans carrying Mobile Medical Units in high-risk areas.
- Ensure adequate medical supplies are available at all hospitals and health centres.
- During a heat alert, generate weekly report of health impact of heat wave for the RMC Nodal Officer.
- If required, rope in medical interns of allopathic, ayurvedic and homoeopathic medical colleges to assist health service providers in coping with the increased caseload during a heat wave.
- Monitor outreach work of primary health centres and ASHA workers during a heat alert.
- Checking and cross-checking emergency preparedness of primary health centres.
- Issues directions to Primary Health Centres, nursing homes, private and govt hospitals to report daily heat morbidity and mortality for monitoring the cases.

RMC Labour and Employment Department

- Encourage employers to shift outdoor workers' schedules away from the peak afternoon hours (1pm-5pm) during a heat alert.

- Arrange for emergency ice-packs and protective gear like cooling vests to traffic police, bus drivers-conductors and construction workers.

108 Emergency Service

- Ensure adequate supply of ice packs and IV fluids.
- During a heat alert, disseminate text messages to warn residents in high-risk areas.

7.10.3 Roles and responsibilities in Phase 3: Post-summer (July to October)

RMC Nodal Officer

- Organize annual heat action plan evaluation meeting with representatives from key government departments and other stakeholders.
- Evaluate the plan based on performance and revise accordingly.
- Upload the revised plan on the RMC website for stakeholders' feedback.
- Undertake tree-plantation in heat hotspot areas. Incentivize builders to plant trees.
- Establish cool resting centres in high-risk areas of the city.

RMC Health Department

- Perform an epidemiological case review of ward level heat-related morbidity and mortalities during summer.
- Based on average daily temperatures, gather epidemiological data on heat risk factors, illness and death.
- Incorporate data and findings into future versions of the Heat Action Plan.
- Measure mortality and morbidity rates based on data before and after the plan's interventions.

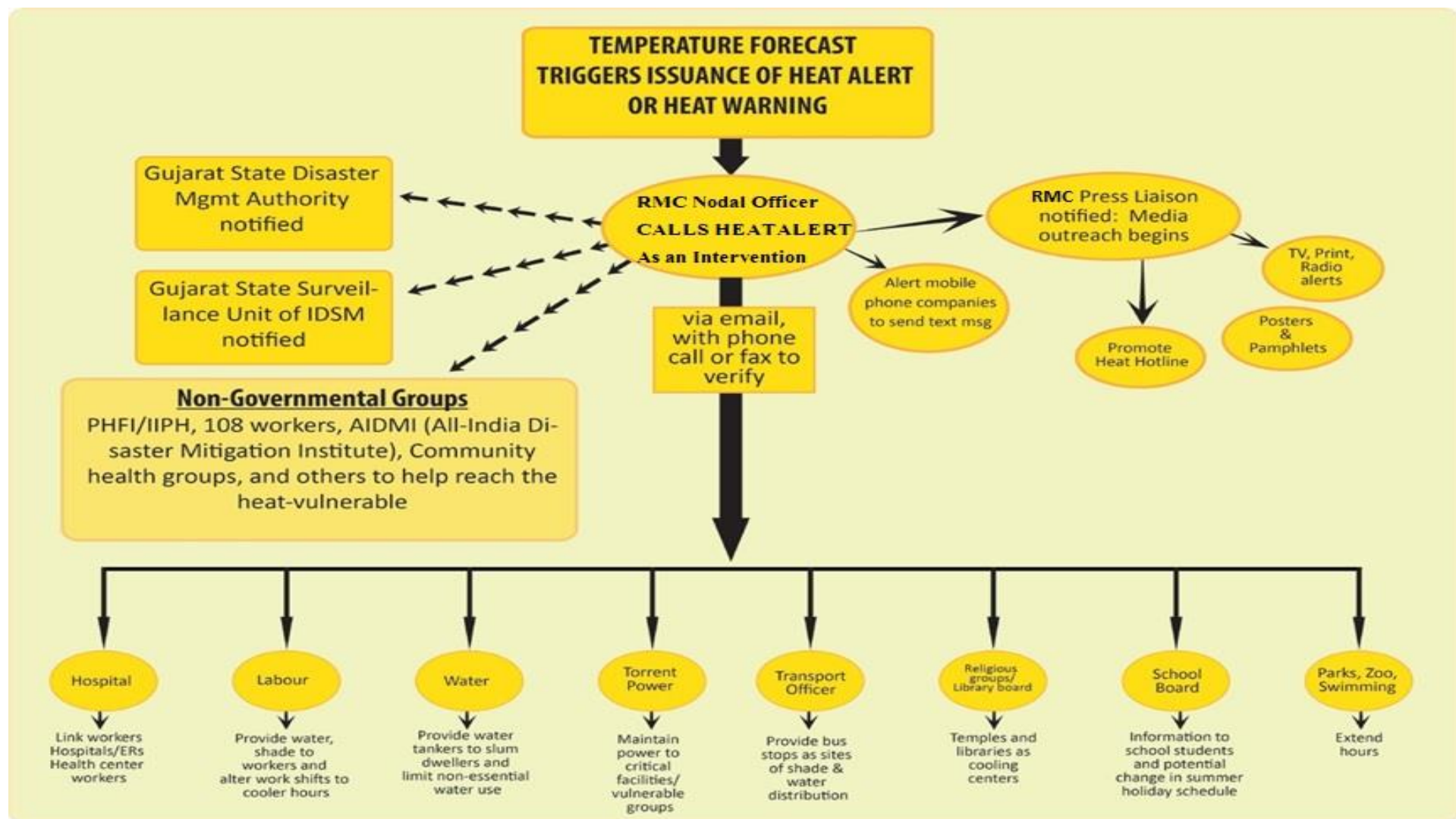


Figure: Communication plan when RMC nodal officer releases a heat alert

7.11 Brief description of what to do when for key service providers

Public health managers

Pre-summer

- ✓ Identify vulnerable areas
- ✓ Ensure adequate inventories of medical supplies in health centres
- ✓ Arrange for additional paramedics and clinicians
- ✓ Identify cooling centres and barriers to accessing cooling centres

During heat event

- ✓ Prepare rapid response team
- ✓ Distribute pamphlets with “Dos and Don’ts” among vulnerable communities
- ✓ Effectively send a “Take Care but Don’t Panic!” message to community
- ✓ Arrange for vans carrying Mobile Medical Units for deployment in the high-risk zone
- ✓ Ensure additional medical vans are available during red alerts

Post-summer evaluation

- ✓ Participate in annual evaluation of heat action plan evaluation
- ✓ Review revised heat action plan and suggest needed

Primary health centres and community health workers

Pre-summer

- ✓ Advise community on treatment and prevention of heat related illness
- ✓ Sensitize and train frontline workers
- ✓ Develop and execute school health programmes with support from Department of Education
- ✓ Create awareness campaigns in slum communities
- ✓ Coordinate with non-profits for community outreach

During heat event

- ✓ Re-check stocks
- ✓ Ensure primary health centre’s preparedness to respond to emergencies
- ✓ Visit at-risk populations for monitoring and prevention
- ✓ Communicate information on tertiary care and 108 service

Post-summer evaluation

- ✓ Participate in annual evaluation of heat action plan
- ✓ Review revised heat action plan and recommend needed amends

RMC Press Officer

Pre-summer

- ✓ Secure commercial airtime slots for health advisories and public service announcements
- ✓ Identify public areas to display health alerts during summer
- ✓ Organize training of community health workers and medical professionals
- ✓ Activate heat related telephone-hotlines
- ✓ Place temperature forecasts in newspapers
- ✓ Install LED screens with scrolling temperature data

During heat event

- ✓ Issue heat-related health warnings in the media
- ✓ Contact local FM radio and TV stations for health and weather advisories
- ✓ Release advisories through SMS and WhatsApp platforms using centralized mobile databases
- ✓ Contact transport department to place warnings on buses

Post-summer evaluation

- ✓ Evaluate efficacy of advocacy and campaign outreach and other communications
- ✓ Participate in annual evaluation of heat action plan
- ✓ Review revised heat action plan and suggest amends

Labour Department

Pre-summer

- ✓ Organize orientation for factory medical officers and general practitioners on health effects of heat stress or exposure
- ✓ Compile list of factory medical officers and contractors for heat action communications from Nodal Officer
- ✓ Prepare outreach and advocacy strategy for unorganized labour
- ✓ Use maps of construction sites to identify high-risk outdoor workers
- ✓ Conduct advocacy campaigns in high-risk areas

During summer

- ✓ Ensure water supply at work sites
- ✓ Request use of air-conditioners at factory facilities
- ✓ Extend work hours of Occupational Health Centres
- ✓ Consider long afternoon break or modify working hours to avoid heat exposure
- ✓ Arrange for emergency ice-packs and preventive gear like cooling vests to traffic police, bus drivers-conductors and construction workers

Post-summer evaluation

- ✓ Participate in annual evaluation of heat action plan
- ✓ Review heat action plan and suggest amends

108 Emergency Service

Pre-summer

- ✓ Prepare handouts for paramedics on heat related illnesses
- ✓ Uses informative visuals on ambulances to build public awareness
- ✓ Establish dynamic strategic deployment plan for ambulances
- ✓ Ensure adequate supply of IV fluids
- ✓ Identify at-risk areas

During summer

- ✓ Ensure adequate staff and stock of required medicine and equipment
- ✓ Keep accurate record of pre-hospital care
- ✓ Send messages to 108 Emergency Service employees on heat action plan and heat alerts
- ✓ Activate dynamic strategic deployment plan for the ambulance service

Post-summer evaluation

- ✓ Provide data to RMC steering committee
- ✓ Participate in annual evaluation of heat action plan
- ✓ Review revised heat action plan and recommend amends

7.12 Key actions for summer - Rajkot



Awareness Campaigns

1. Hoardings, posters, to be displayed by city LED TVs at various locations, distribution of pamphlets.
2. Awareness workshops for those whose jobs require sun exposure – traffic police, hawkers, street vendors, construction workers, outdoor and indoor women workers.
3. Awareness campaigns need to focus on vulnerable groups – pregnant women, infants, children, senior citizens and people with co-morbidities.
4. Information about medical facilities and nearest hospitals with cooling wards should be provided to the citizens.
5. Schools should be provided with material on heat stress management and also the remedial actions for managing heat stress should be communicated.
6. Storing cold water, ORS and cool packs in schools and school buses/vans/rickshaws.



Mitigation Measures

1. Keeping open gardens, cooling shelters and other possible cooling centres with ventilation, and ensuring availability of water.
2. Making water and sheds available at open construction sites.
3. Pilot project on roof painting with white colour for a cool roof or distribution of gunny bags for placing on tin roofs/asbestos – keeping these drenched will lower the room temperature.
4. Provision of water points and ORS at construction sites, bus stands and other public places during political processions and rallies in summer.
5. Distribution of cooling vests to on-duty traffic police personnel
6. Water tanker to be made available on priority call in slums during orange/red alert days.



Early Warning Communication

1. SMS and WhatsApp messages for early warning to citizens, NGOs, citizen welfare groups, construction contractors and school authorities
2. Public announcement through microphones across the city during orange and red alert days, a day prior to the forecasted day.
3. Press releases and campaigns on radio, TV, newspapers and websites



Medical Preparedness

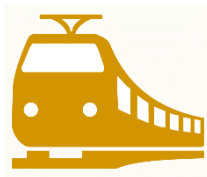
1. Primary health centres' preparedness should include setting up of a helpline.
2. House-to-house visits by Asha/ANM to identify vulnerable groups viz. pregnant women, infants, children, senior citizens and people with co-morbidities. Routine visits to these groups should be carried out.
3. Stocking ORS and cool packs at the health centres and readiness with re-hydration as well as shock management treatments.
4. Medical camps on day of red alerts at hotspots



Monitoring and Analysis

1. Ward-wise recording of heat stroke cases, cause of death and monitoring daily mortality and daily hospital admission due to all causes and heat-related causes
2. Monitoring and analysis of the morning temperatures recorded at automatic weather stations and issue early warnings

7.12.1 Role of Departments



Railways

1. Provision of cool/ AC waiting rooms at railway stations
2. Installation and proper maintenance of the fans and cold water points
3. Stocking ORS and cool packs, and making available re-hydration as well as shock management treatments
4. Availability of medical staff at railway stations
5. Frequent and updated announcement on temperature and mitigation measures and availability of medical care at the station
6. Posters and stickers on notice boards, along with display on LED TVs



Transport (bus service)

1. ORS, first-aid kits and water points to be made available at bus terminals and buses
2. Awareness posters/stickers in buses and bus stops
3. Announcement of temperature and mitigation measures on bus display screen



Health Department

1. Installation of ice-pack dispensers for easy access by vulnerable communities
2. Regular update of admissions and emergency case records to track heat-related morbidity
3. Adopt heat-focused examination procedures at hospitals and health centres
4. Enhance targeted training programmes, capacity building efforts and communication on heat-related illness for medical staff at city hospitals and health centres, including nursing staff, paramedics, and frontline workers.
5. Well-equipped primary health centres, 108 emergency centres, ambulances and hospitals for the treatment of heat-related illnesses



Labour and Employment Department

1. Identify high-risk outdoor workers and conduct outreach and advocacy.
2. Conduct awareness campaigns in hotspot areas and among vulnerable groups along with the distribution of pamphlets and posters
3. Organize training for employers, outdoor labourers and workers on the health impacts of extreme heat
4. Provide information on mitigation and adaptive measures to prevent heat exposure, heat stress and associated debility



108 Emergency Service

1. Put up displays on ambulances to build public awareness.
2. Identify vulnerable populations in hotspot areas and be in a state of readiness to reach immediate relief in response to calls reporting illnesses.



Schools

1. Conduct child-friendly sessions on heat stress management focusing on preventive measures, like avoiding exposure and staying hydrated.
2. Stock ORS, first-aid kits and cool water for heat stress management.
3. Equip with resources for heat protection.
4. Arrange for a 'Teach the Teachers' workshop.

Since the impact of heat stress management has to be created at the ward level it is good to know the community based groups that exist in a ward. These are:

- **Heat Stress Management Committee** – comprises ward councilors, health officials, ward officials of the municipal corporation, representatives of NGOs and of research institutions working at the ward level.
- **School Management Committee** – this community based committee was introduced by the Right to Education Act of 2009 and includes parents, school principal, a teacher and an elected representative of the ward.
- **Mahila Arogya Samiti** – this 10-member committee of women in the slum already works in coordination with the municipal corporation, and covers 50 to 100 households in a slum. This group is facilitated by ASHA, holds monthly meetings and works to achieve health related goals.

These community groups can be empowered to take a leadership role in orchestrating what needs to be done in the ward during heat waves. With proper facilitation and coordination by each of the six key departments as mentioned in the previous chapter:

- Drinking water and sanitation
- Education
- Electricity
- Transport
- Labour
- Health and the National Urban Health Mission
- Women and child development department that runs the child development (Aanganwadi) centres in the community

Heat Stress Management Committee, School Management Committee, Mahila Arogya Samiti groups can prove to be an asset in the mitigation of heat stress at the ward level. Ward level intervention framework will be further strengthened through public awareness and capacity building of ward level stakeholders.

Rajkot Heat Wave Action Plan through its climate adaptive, targeted and focussed approach envisions to achieve Zero mortality to heat waves and enhance heat resilience of Rajkot city.

Annexure

Template for daily Heat Mortality/Morbidity data collection

Name of the Urban Primary Health Centre/ hospital									
Ward no-									
	No of cases	Gender			Pregnant women if any	Age Group			
		No of male	No of Female	Other		0-14	15-35	35-60	60 Above
Heat Cramps									
Heat Exhaustion									
Heat Stroke									
Mortality due to heat stroke									

રાજકોટ ક્લાઈમેટ એક્શન પ્રોજેક્ટ

રાજકોટ હીટ એક્શન પ્લાન

લૂ ઝુલવેણ નીવડી શકે છે, પણ તેનાથી બચવું શક્ય છે.
લૂ થી બચવા માટેના ઉપાયો



વારંવાર પાણી પીવું



તડકામાં ખાસ કરીને બપોરના
૧:૩૦ થી ૩:૩૦ ના ગાળામાં
બહારજવાનું ટાળો



આછા/સફેદ રંગના ખુલા
કપડા પહેરવા



બાળકો, ઘરડા અને ગર્ભવતી
મહિલાઓનું ખાસ ધ્યાન રાખવું



ઠંડક આપે તેવા પીણા જેમકે
ઓ. આર. એસ. છાસ,
જ્યુસ, શરબત, શિકંજુ
નું સેવન કરવું



અગર ઘરની બહાર છે
તો છાંયડામાં રહેવાનું રાખો



ઘરના છાપરાને ચૂનો/સફેદ
રંગથી પેન્ટ કરવું.



માથા પર ભીનું કપડું
અથવા શરીરને
કપડાથી ઢાંકીને બહાર જવું



ગરમી માં ભારે
શારીરિક
પ્રવૃત્તિઓ ટાળો



કાંડા પર ઠંડુ પાણી રેડવું
જેથી શરીરનું
તાપમાન ઓછું રહે



પાર્ક કરેલી કારમાં
બાળક/પાલતુ જાનવર
ને છોડશો નહીં

લૂ ના લક્ષણો



શરીરના તાપમાનમાં વધારો
શ્વાસ લેવામાં થવો પણ પરસેવો ન છુટવો
મુશ્કેલી થવી



માથાનો દુખાવો અથવા
માથુ ભારે લાગવું



ચામડી શુષ્ક અને
લાલ થવી



ઉલ્ટી થવી



બેભાન થઈ જવું



સ્નાયુઓમાં તણાવ

લૂ માટેની પ્રાથમિક સારવાર

(1) વ્યક્તિને ઠંડા અથવા છાંયડો હોય એવી જગ્યાએ લઈ જવું

(2) નજીકના આરોગ્ય કેન્દ્રમાં લઈ જવું અથવા

એમ્બ્યુલન્સને ફોન કરવો(૧૦૮)

(5)

બની શકે તેટલા ટીલા
અને પાતળા કપડાં પહેરવા

(4)

અગર બેભાન ના હોય
તો ઠંડુપાણી પીવડાવવું



(8)

શરીરને ઠંડુ રાખવા ઠંડી હવા
આપે તેવા ઉપકરણોનો ઉપયોગ કરવો

(6)

શરીર પર
પાણી છાંટવું

(3)

વ્યક્તિને પગ ઉપરની
તરફ રાખીને સુવડાવવું

(7) ભીનું કપડું રાખવું



गर्मी/लू के प्रकोप से बचाव

15th Integrated Research and
IRADe Action for Development

गैल्टो अलर्ट
सांतापि अधिकतम तापमान
34°C की संभावना
अधिकतम तापमान 37°C से
34°C तक बढ़ेगा

ऑरेंज अलर्ट
सांतापि अधिकतम तापमान
36°C से संभावना
अधिकतम तापमान 37°C से
34°C तक बढ़ेगा

रेड अलर्ट
सांतापि अधिकतम तापमान
37°C से संभावना
अधिकतम तापमान 37°C से
34°C तक बढ़ेगा

अत्यधिक संवेदनशील समूह



गर्मी/लू से बचाव के उपाय

4-5 लीटर
ठंडा पानी पीएं

खाद्य, शिकंजी,
ORS बार बार पीएं

धीले-ढाले सूती
कपड़े पहनें

ठंडे जल से
स्नान करें

दोपहर 12 बजे से 4 बजे तक
बाहर न निकलें

छत पर घूमे का पेंट करें या पीले
खोरी का उपयोग करें

अपनी कलाई को ठंडे पानी
के नीचे रखें

छाया में बैठें

सिर और चेहरा ढककर रखें
और छाया लेकर चलें

अव्यक्त शारीरिक
अवस्था में बनें

बच्चों या पालतू जानवरों को
पार्क की गर्म कार में न छोड़ें

खिदकिचो/दरवाजे पर
शीले पर्दे लटकाने

गर्मी/लू लगने के लक्षण

गर्मी से बेहोशी

- सिर चकराना
- बेहोशी
- चक्कर आना
- मूर्छा

गर्मी से थकान

- बेहद कभी महसूस होना
- सिर चकराना, सिरदर्द
- उबकाई
- पसीना और प्यास
- हाटौर का तापमान बढ़ जाना
- नींद-चेष्टियों में ऐंठन
- घबराहट, चक्कर

लू लगना/ हीट स्ट्रोक

- तेज सिरदर्द
- सीस लेने में कठिनाई
- तेज नब्ब
- उबकाई और उल्टी
- खुरक जल त्वचा
- बुखार बढ़ने पर पसीना न आना
- बुखार 104°F से ऊपर, जोकि पैरासिटामोल/इबुप्रोफेन से कम नहीं हो रहा
- बेहोशी

फर्स्ट एड

- 1 व्यक्ति को अंदर ले जाएं और पंखे/कूलर/AC कम में रखें
- 2 तेजी को नीचे लिटाएं और उसकी टांगें और खुंई को ऊंचा रखें
- 3 कपड़े ढीले करें
- 4 हाटौर को एक पीले चादर से ढकें
- 5 बर्फ और ठंडे पानी से पोंछें
- 6 यदि साफ है तो थोड़ा थोड़ा ठंडा पानी पिलाएं
- 7 प्रचुर मात्रा में आईसी प्रलुप्त और इलेक्ट्रोलाइट दें
- 8 फोन जरूरत में माली करवाएं अगर बुखार 104°F से ऊपर है, जोकि पैरासिटामोल/इबुप्रोफेन से कम नहीं हो रहा



102 आपात चिकित्सा
सहायता हेतु कॉल करें

Heat Health Advisory



Medical Practitioners Training Workshop- Rajkot- May 2019



Interactive Stakeholders Workshop- Rajkot- June 2019



Dissemination of Heat Stress Advisory among 15,000 plus people in Rajkot



9 LED Screen at different place of Rajkot City



5263 Stickers at different place of City



Water Points and Buttermilk distribution at more than 300 locations



ORS corner and Rapid Response Team (RRT) at all Urban Health Centres (UHC)

RMC to lower heat wave warnings by half a degree

Nimesh Khakhariya | TNN | Updated: Jun 28, 2019, 11:51 IST

✉ 🖨 A- A+



A workshop on heat stress action plan was held on Wednesday

when average temperature touches 41°C. But RMC will declare yellow alert at 40.5°C itself.

RAJKOT: With temperatures gradually rising over the years, and in fact, the past one decade has registered the historic maximum average temperature, the Rajkot Municipal Corporation (RMC) has decided to revise its warning system to reduce heat casualty. Therefore, it will lower the standard heat wave alerts by half a degree. For example, yellow alert is issued

Policy Level Intervention- City Government (Gujarat, India) Revised warning system by - lowering the heat alert by 0.5°C - The current threshold is 40.5°C reduced from 41°C



રાજકોટ મહાનગરપાલિકા આરોગ્ય શાખા

સ્વચ્છ
સર્વેક્ષણ
2019

તારીખ: ૨૫/૦૪/૨૦૧૯

:: અખબારી યાદી ::

ભારતીય હવામાન વિભાગ (IMD) દ્વારા
આગામી દિવસોમાં રાજકોટ શહેરમાં ઓરિજન
એલર્ટ જાહેર

હિટવેવથી બચવા રાજકોટ મહાનગર પાલિકા દ્વારા
3P નો કન્સેપ્ટ (Concept) : પ્રોટેક્શન (રક્ષણ),
પ્રિફોશન (સાવચેતી), પ્લેન્ટી ફલુઇડ (વધુ પ્રવાહી)

ભારતીય હવામાન વિભાગ (IMD) દ્વારા આગામી દિવાસો માટે ઓરિજન એલર્ટ (વધુ ગરમ-હીટ એલર્ટ ડે) જાહેર કરેલ છે. જ્યારે વાતવરણનું તાપમાન ૪૩.૩ સે. થી ૪૪.૯ સે. થાય ત્યારે ઓરિજન એલર્ટ જાહેર કરવામાં આવે છે.

ઓરિજન એલર્ટ દરમિયાન વધતી ગરમીના કારણે હીટ સ્ટ્રોક (થકાવટ), હીટ કેમ્પ્સ (રનાયુમાં દુઃખાવો અને ખેંચાણ) માથાનો દુઃખાવો, નબળાઈ, હીટ સીન્કોપ - કોઈ ખેંચ કે સ્ટ્રોકની બીમારી ના હોવા છતાં બેભાન થઈ જવું, બદલાયેલી માનસિક સ્થિતિ વગેરે થવાની શક્યતા છે.

હિટવેવની અસરથી શરીરમાં પાણીનું પ્રમાણ ઘટવાથી (ડીહાઇડ્રેશન) થી માથું દુઃખવું, હૃદયના ઘબકારા વધી જવા, ચામડી સુકી થઈ જવી, શરીરનું તાપમાન વધી જવું જેવા ચિહ્નો જણાય છે જે ગંભીર નિશાની છે. રાજકોટ મહાનગરપાલિકા દ્વારા ઓરિજન એલર્ટને અનુલક્ષીને 3P નો કન્સેપ્ટ (Concept) જાહેર કરેલ છે. P=પ્રોટેક્શન (રક્ષણ), P= પ્રિફોશન (સાવચેતી), P=પ્લેન્ટી ફલુઇડ (વધુ પ્રવાહી)

1. P = પ્રોટેક્શન (રક્ષણ) :

- દિવસ દરમિયાન અનિવાર્ય સંજોગો સિવાય તથા ખાસ કરીને બપોરના ૧૨ થી ૪ સમય દરમિયાન બહાર જવાનું ટાળવું.
- દિવસ દરમિયાન સુતરાઉ, આછા કલરના કલરના પહોળા કપડા પહેરવા.
- બહાર નીકળવાનું થાય ત્યારે ગોગલ્સ, છત્રી, કેપ, તથા સનસ્ક્રીન લોશનનો ઉપયોગ કરવો.
- બહાર જતી વખતે પીવાના પાણીની બોટલ અવશ્ય સાથે રાખવી.

2. P = પ્રિફોશન (સાવચેતી) :

- સગર્ભા માતા, નાના બાળકો, વૃદ્ધો, મેદસ્વી લોકો, તથા લાંબી બીમારીથી પીડિત લોકોએ ખાસ તકેદારી રાખવી
- અશક્ત તથા હૃદય, કીડની અને કેન્સર જેવા રોગના દર્દીઓ એ સાવચેતી રાખવી.
- વધારે શ્રમ પડે તેવી કામગીરી ન કરવી.
- દિવસ દરમિયાન શ્રમિકો તથા સતત તડકામાં કામ કરતા લોકોએ થોડા થોડા સમયે છાંયડા તથા ઠંડક વાળી જગ્યાએ આરામ કરવો. કામકાજની જગ્યા એ સ્વચ્છ અને પીવા લાયક પાણીની પુરતી વ્યવસ્થા કરવી.
- દિવસ દરમિયાન બંધ વાહનોમાં મુસાફરી કરવાનું ટાળવું.
- ઘરમાં સ્ત્રીઓએ બપોરની રસોઈ બનાવવાનો સમય વહેલો કરવો તેમજ રસોઈ બનાવતી વખતે હવાની અવર જવર રહે તે માટે બારી-દરવાજા ખુલા રાખવા.
- યા -કોફી કે અન્ય કેફી પીણાનો ઉપયોગ ટાળવો.
- દૂધ તથા માવાની મીઠાઈ તેમજ અન્ય ભારે ખોરાક નો ઉપયોગ ટાળવો.
- વાસી, તળેલો, ફરસાણ, કે વધુ પ્રોટીન વાળો ખોરાક ટાળવો.

3. P = પ્લેન્ટી ફલુઇડ (વધુ પ્રવાહી) :

- શરીરમાં ગરમીના કારણે ડીહાઇડ્રેશન થતું રોકવા પુષ્કળ માત્રામાં પ્રવાહી લેવું.
- દિવસ દરમિયાન તરસ લાગે તેનાથી વધારે શુદ્ધ ૧૦ થી ૧૨ ગ્લાસ પાણી પીવું જોઈએ.
- શરીરને ઠંડક આપે એવા પીણા લેવા જેવા કે છાસ, શીકન્ઝ, લીંબુ શરબત, પરીયાળી શરબત, નારિયેળ પાણી વગેરેનો ઉપયોગ કરવો.
- તરબુચ, શકરટેટી, મોસંબી, સંતરા, દ્રાક્ષ, જેવા જ્યુસી એટલે કે રસવાળા ફળોનો ઉપયોગ વધારવો.
- લૂ લાગવાના ચિહ્નો જણાય તો ORS (ઓરલ રીહાઇડ્રેશન સચેટ) એટલે કે (ખાંડ+મીઠું)ના દ્રાવણનો ઉપયોગ કરવો.
- ઘરગથ્થું ORS (ઓરલ રીહાઇડ્રેશન સચેટ) બનાવવાની રીત
 - ✓ ૬ ચમચી ખાંડને ૧ લીટર શુદ્ધ પીવા લાયક પાણીમાં ઓગાળવી.
 - ✓ આ દ્રાવણમાં ૧ ચમચી મીઠું (નમક) ઉમેરવું.
 - ✓ હવે આ દ્રાવણને પીવા માટે ઉપયોગ કરી શકાય.

આરોગ્ય અધિકારી
રાજકોટ મહાનગરપાલિકા

અખબારી યાદી

હિટ વેવને પગલે નાગરિકો જોગ
રાજકોટ મહાનગરપાલિકાની માર્ગદર્શિકા

ઉંચા તાપમાનમાં તકેદારી રાખવા મ્યુનિ. કમિશનર શ્રી
બંછાનિધિ પાનીની અપીલ

હિટવેવ (ગરમીનું મોજું) એ અતિશય ઉષ્ણ આબોહવાનો સમયગાળો છે. IMD (ભારતીય હવામાન વિભાગ) મુજબ જ્યારે હવાનું તાપમાન સદા પ્રદેશોમાં ઓછામાં ઓછું ૪૦ સે. (૧૦૪ ફે.) થાય અને પર્વવીચ પ્રદેશોમાં ૩૦ સે. (૮૬ ફે.) થી વધારે ત્યારે ગરમીનું મોજું (હિટવેવ) ગણાય છે. IMD (ભારતીય હવામાન વિભાગ) દ્વારા હિટવેવની સમજ તથા ચેતવણી માટે ગરમીના મોજા માટેના રંગ સંકેતો (કલર એલર્ટ) નક્કી કરેલ છે જે મુજબ

વ્હાઈટ એલર્ટ	૪૧ સે. થી ઓછું	સલામત
યેલો એલર્ટ	૪૧.૧ થી ૪૩.૩ સે.	ગરમ (Hot Day)
ઓરેન્જ એલર્ટ	૪૩.૩ થી ૪૪.૯ સે.	વધુ ગરમ (હીટ એલર્ટ ડે)
રેડ એલર્ટ	૪૫ સે. થી વધારે	અત્યંત ગરમ

❖ સૌથી વધુ અસર

- નાના બાળકો અને વૃદ્ધો
- એકલા રહેતા લોકો, સામાજિક રીતે અલગ રહેતા લોકો, માનસિક બીમાર વ્યક્તિઓ
- હૃદય, કીડની, કેન્સર ફેફસાની લાંબી બીમારીથી પીડિત લોકો
- અશક્ત લોકો

❖ ગરમીને કારણે માંદગીના ચિન્હો અને મૃત્યુના વધારાના કારણો

- હીટ સ્ટ્રોક - ગરમીથી થકાવટ - હીટ કેમ્પ્સ
- ગરમીની ફોલ્લીઓ : ગરમીમાં બહાર ફરવાથી કે પરસેવાથી થાય
- હીટ કેમ્પ્સ : ગરમીમાં કામ કરવાથી સ્નાયુઓનું દુઃખાવા સાથે ખેંચાણ
- ગરમીમાં થકાવટ : માથાના દુઃખાવા સાથે સામાન્ય નબળાઈ, થકાવટ
- હીટ સીન્કોપ : કોઈ સીઝર કે સ્ટ્રોકની બીમારી ન હોવા છતાં બેભાન થઇ જવું
- હીટ સ્ટ્રોક : બદલાયેલી માનસિક સ્થિતિ (દિશાહીનતા, ચિત્તભ્રમણા, ખેંચ વગેરે) જેમાં શરીરનું તાપમાન ૪૦°C કે તેથી વધુ હોય અને સ્ટ્રોક, ચેપ કે દવાના ઓવરડોઝના કોઈ ચિન્હો ન હોય.

❖ સારવાર :-

- દર્દીને પર્યાવરણીય ગરમીના સંસર્ગમાંથી દૂર કરો અને શારીરિક પ્રવૃત્તિ બંધ કરાવો.
- બહારથી ઠંડક આપવાની પ્રક્રિયા શરૂ કરો.
- ઠંડા પાણીમાં ભીનો કરેલો ટુવાલ કે આઈસ પેક બગલ, જાંઘ અને ગળા ફરતે રાખો; જો દર્દીની સ્થિતિ સ્થિર હોય તો ઠંડા પાણીના કુવારાથી સ્નાન પણ કરી શકે છે.
- ઠંડા પાણીનો સ્પ્રે આપી શકાય.
- ભીની ચામડી પર ઠંડી હવા છોડવા માટે પંખાનો ઉપયોગ કરો.
- શરીરનું તાપમાન ૪૦°C થી ઓછું હોય તો, દર ૫ મિનિટે ફરીથી ચકાસો; સુધારો જણાય તો મોંઢેથી પ્રવાહી (ઓ.આર.એસ., ઠંડા પ્રવાહી) આપવાનું શરૂ કરો અને અવલોકન ચાલુ રાખો.
- શરીરનું તાપમાન ૪૦°C કે તેથી વધુ હોય તો, IV (નસમાં) સારવાર શરૂ કરી તુરત જ ઈમરજન્સી વિભાગમાં વધુ સારવાર માટે રીફર કરો.